Learning to Anticipate Flexible Choices in Multiple Criteria Decision-Making Under Uncertainty

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2 authors:

Carlos R. B. Azevedo
Bain & Company

Fernando J. Von Zuben
State University of Campinas (UNICAMP)

Comments to Editor-in-Chief regarding the review of CYB-E-2014-07-0677.R1

Carlos R. B. Azevedo, *Member, IEEE* and Fernando J. Von Zuben, *Senior Member, IEEE*

Dear Prof. Jun Wang, Editor-in-Chief, and Associate Editor: we have carefully performed the requested minor changes for the publication of the manuscript entitled "**Learning to anticipate flexible choices in multiple criteria decision-making under uncertainty**".

I. Minor Changes Requested by Reviewers

Reviewer 1 pointed out a potential mistake in Fig. 11(d). The labeling of Fig. 11(d) was in fact correct and, actually, the error was in the text. The sentence highlighted by the reviewer now correctly reads as "*For the HSI, the RDM strategy outperformed MDM"*. In addition, two minor typos identified by the reviewer were corrected on pages 2 and 10. The Reviewer 1 and the remaining reviewers have made no further requests.

A. Additional Minor Changes

In addition to the changes requested by the reviewer, we have included a link to the source code in C++ for our implementation of the ASMS-EMOA algorithm (see footnote 3 at page 6 of the main text), proposed in this paper, in order to make it easier for other researchers to reproduce our results.

Also, we have updated both the Abstract and the first paragraph of the Introduction, with the goal of making the contributions of this paper even more clear.

Finally, we have removed a couple of redundant sentences from the last paragraphs of the Future Works subsection and added an Acknowledgment section.

B. Updated References

Please note that we have updated reference [26] of our previous submission to a more recent and extended version of the study of Salomon et al., which appeared in the issue 11, vol. 44 IEEE Transaction on Cybernetics on Nov. 2014 (reference [25] in the current revision). We now cite a total of three relevant papers which were published in this journal (see references [4,5,25]).

Acknowledgment

We thank the Editors and reviewers for the careful assessment of our research.

Learning to Anticipate Flexible Choices in Multiple Criteria Decision-Making Under Uncertainty

Carlos R. B. Azevedo, *Member, IEEE* and Fernando J. Von Zuben, *Senior Member, IEEE*

***Abstract*—In several applications, a solution must be selected from a set of trade-off alternatives for operating in dynamic and noisy environments. In this paper, such multi-criteria decision process is handled by anticipating flexible options predicted to improve the decision maker future freedom of action. A methodology is then proposed for predicting trade-off sets of maximal hypervolume, where a multi-objective metaheuristic was augmented with a Kalman filter and a dynamical Dirichlet model for tracking and predicting flexible solutions. The method identified decisions that were shown to improve the future hypervolume of trade-off investment portfolio sets for out-of-sample stock data, when compared to a myopic strategy. Anticipating flexible portfolios was a superior strategy for smoother changing artificial and real-world scenarios, when compared to always implementing the decision of median risk and to randomly selecting a portfolio from the evolved anticipatory stochastic Pareto frontier, whereas the median choice strategy performed better for abruptly changing markets. Correlations between the portfolio compositions and future hypervolume were also observed.**

***Index Terms*—Anticipatory learning; Bayesian tracking; multi-objective optimization; hypervolume-based decision making.**

I. Introduction

PRESSURE to act under incomplete information may often result in unintended consequences. Nevertheless, when there is room for revising decisions, implementing flexible solutions can avoid shortage of future options, while mitigating the risks of premature commitments to uncertain alternatives. Strategically, promoting flexibility in decision-making can increase adaptability to upcoming changes [1], allowing for greater future freedom of action. This paper therefore contributes with models and algorithms for flexible Multiple Criteria Decision Making (MCDM), interpreting flexibility as a means to robustly postpone preferences specifications.

In a posteriori MCDM, a set of trade-off alternatives is usually searched for by a Multi-Objective Optimization (MOO) algorithm. The Decision Maker (DM) does not interfere until he/she is asked to identify the most suitable alternative. Evolutionary MOO (EMOO) may aid MCDM when preferences are undefined, by using the Pareto Dominance (PD) for ranking alternatives and approximating the Pareto Frontier (PF) with a set of mutually Pareto-incomparable alternatives that can be presented to the DM. In this step, the classical approach is to aggregate the decision criteria into a utility function. Preference elicitation is nevertheless prone to misspecification. For instance, the DM may not be able to rank every pair

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C.R.B. Azevedo and F.J. Von Zuben are with the School of Electrical and Computer Engineering, UNICAMP, Campinas, São Paulo, Brazil. E-mails: {azevedo,vonzuben}@dca.fee.unicamp.br

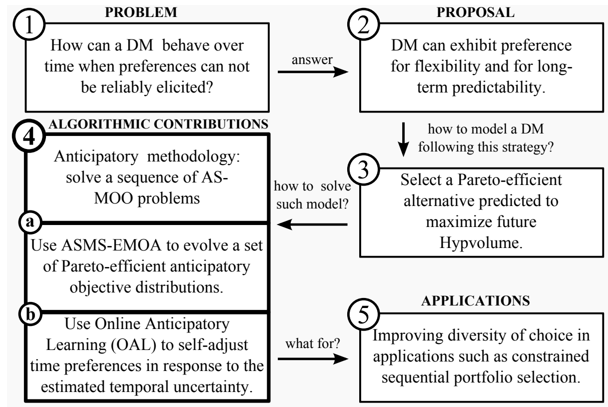


Fig. 1. Research roadmap showing the paper's main goal and contributions.

of alternatives [2], or the rankings may violate transitivity, preventing preferences to be represented with utility functions. Still, the premise in MCDM is that preferences can always be elicited after the DM examines a set of alternatives (see [3]). This is however unrealistic when it is unclear how the trade-offs between the decision criteria change over time and when choice consequences are uncertain.

Handling dynamic and stochastic MOO problems is challenging because, while Pareto-optimal alternatives can be identified in retrospect, their optimality for upcoming data is unknown. To the best of our knowledge, despite existing MCDM/EMOO approaches that can either cope with noisy (e.g. [4]) *or* dynamic conditions (e.g. [5]), there are no EMOO algorithms for simultaneously coping with noisy and time-varying objective functions. In a recent survey, Gutjahr and Pichler [6] concluded that dynamic stochastic MOO is *"still widely unexplored, even on the modeling level. (...)"*.

Therefore, when preferences cannot be reliably specified due to non-stationarity and noise, we propose modeling a DM whose goal becomes achieving future improved flexible and predictable sets of alternatives to choose from in later stages. In order to improve such a DM future options, we propose maximizing set-based quality indicators (see [7]) over time, among which we argue that the Hypervolume (Hypv, $\mathcal{S}$) [7] is consistent with our notion of flexibility.

We then propose an online EMOO/MCDM methodology to anticipate flexible choices, enabling the DM to access future option sets that match more diverse ranges of admissible choices. Otherwise, if the DM compromises for one path and later gravitates toward other, the cost to adapt could be very high. The dilemma of disclosing preferences under uncertainty is thus alleviated by postponing their specification in favor of future freedom of choice. The proposal can be properly formalized into the five steps of Fig. 1.

It is natural to assess our methodology in Portfolio Optimization (PO) because financial markets invite handling sources of substantial uncertainty. PO aims to diversifying capital across investment assets for risk management [8]. Usually, the assets yielding higher returns are those leading to the gravest risks. Composing a portfolio to match the investor attitude to risk is therefore challenging and prone to inconsistencies, which calls for MOO approaches to learning the underlying trade-offs. Kolm et al. [8] suggested that all "*trade-offs between risk-adjusted returns and trading costs*" should be tracked over time. We thus validate the Anticipatory Stochastic MOO (AS-MOO) methodology on three stock indices: London's FTSE-100, Dow Jones Index, and Hong Kong's HSI. Also, we design an artificial PO instance generator, controlling periodicity and severity of change (see Sec. V-C), thus providing different regime switching patterns.

The paper is organized as follows: related works and background on flexibility and its relation with PD and Hypv are provided in Sec. II. The AS-MOO model is presented in Sec. III, whereas Bayesian tracking, OAL, and the Anticipatory $\mathcal{S}$-Metric Selection EMOO algorithm (AMS-EMOA) are formalized in Sec. IV. Section V presents the experimental setup and the portfolio benchmark generator. Finally, Secs. VI and VII provide results and conclusions, respectively.

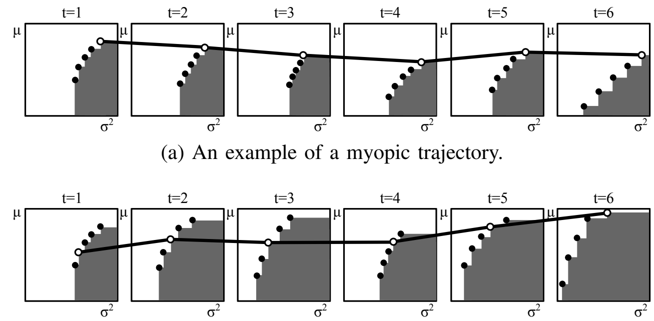
II. Related Work and Background

As far as we are aware, our research is the first to investigate models based on set indicators and EMOO algorithms for simultaneously handling dynamic, cost-adjusted, and noisy objective functions. In this section, we cover related works that handled only subsets of those problem features. Early models have been theoretically analyzed, but not experimentally verified, since the 1970's [9], [10].

Predictive metaheuristics can be traced back to the early 2000's, e.g. [5], [11], [12]. The main drawback of early approaches is the reliance on problem-dependent ad-hoc strategies, besides the inability to directly estimate the future performance of candidate solutions. Stochastic and dynamic single-objective optimization has been tackled e.g. in Bosman and La Poutré [13]. Metaheuristics integrating a Kalman Filter (KF, see Sec. IV-A) were also proposed, e.g. [14], although tracking was mostly used to support ad-hoc search operators.

Improving future freedom of action by means of the causal entropic principle has been recently highlighted as a fundamental building block for general intelligence [15]. Algorithms such as Monte Carlo Tree Search (MCTS) [16] have also benefited from the exploitation of actions predicted to keep options open in multi-objective reinforcement learning.

Axiomatic accounts of preference for flexibility [17] assume that DMs always enjoy enlarging their set of options. A related account [18] regarded essential options as those whose exclusion "*would reduce an agent's freedom*". Essential sets were interpreted as those containing the best options "*once future preferences are well determined*" [19]. Flexibility in a two-stage setting was defined as "*the size of the remaining second-period action set*" [20], whereas theorems asserting that increasing flexibility does not conflict with value and risk optimization were presented in Benjaafar et al. [1]. Finally, Kumar [21] generalized flexibility as "*diversity of choice*" and quantified it using entropic measures.



(a) An example of a myopic trajectory.

(b) An example of an anticipatory maximal Hypv trajectory.

Fig. 2. Example of (a) myopic; and (b) anticipatory sequences of choices in an objective space composed of mean return maximization (μ) and variance (risk) minimization (σ^2). In (a), a myopic DM always choose the most risky Pareto-efficient investment, hoping to maximize near-term wealth. The implemented portfolios influence the future diversity of choice measured by the Hypv (gray area). In (b), a DM anticipating the future maximal Hypv portfolios obtains more efficient and diverse future range of options, and may end up earning higher returns over time by taking safer, flexible choices.

Hypv maximization has been investigated in terms of the distributions of N alternatives over the PF [22], [23]. The goal of Hypv maximization in this paper is to obtain N representative alternatives of all admissible choices. Suboptimal Hypv approximations can leave vast portions of the PF underrepresented. The point of taking decisions to achieve future maximal Hypv distributions of N points is thus to have higher quality approximation of the PF with higher diversity of choice for future preference specifications (see Fig. 2).

III. The Time-Linkage AS-MOO Model

The solution to the AS-MOO model is an approximation of the dynamic Pareto Set (PS) ($\Omega_t^\star|\mathbf{u}_{t-1} \subset \mathcal{R}^d$, i.e., the current PS given that $\mathbf{u}_{t-1} \in \mathcal{R}^d$ was taken[1]), from which one element is identified as the most flexible for improving the DM future range of options. In the following, AS-MOO is described for regimes wherein adaptation costs incur. See Azevedo and Von Zuben [24] for a time-linkage free AS-MOO formulation.

A. Notation and Definitions

We propose representing objective functions in terms of a hidden state $\mathbf{x}_t$, whose transition function $\mathbf{T}$ may depend both on a previous state and on the decision taken at $t-1$:

$$\mathbf{x}_{t+1}|\mathbf{u}_t = \mathbf{T}(\mathbf{x}_t, \mathbf{u}_t, \xi_t) = \mathbf{A}\mathbf{x}_t + \mathbf{B}\mathbf{u}_t + \xi_t, \tag{1}$$

where $\mathbf{A}, \mathbf{B} \in \mathcal{R}^{d\times d}$. Given a decision vector $\mathbf{u}_t \in \Omega_t$, the corresponding objective vector is denoted as $\mathbf{z}_t = \mathbf{f}(\mathbf{u}_t, \mathbf{x}_t, \mathbf{u}_{t-1})$ and is parameterized by the previous decision taken, $\mathbf{u}_{t-1}$. The function $\mathbf{f}$ can be decomposed as

$$\mathbf{z}_t|\mathbf{u}_{t-1} = \mathbf{f}(\mathbf{u}_t, \mathbf{x}_t, \mathbf{u}_{t-1}) = \mathbf{g}(\mathbf{u}_t, \mathbf{x}_t) + \mathbf{h}(\mathbf{u}_t, \mathbf{u}_{t-1}), \tag{2}$$

[1] We sometimes omit the conditional part for convenience: $\Omega_t^\star|\mathbf{u}_{t-1} \equiv \Omega_t^\star$.

where $\mathbf{g}$ encodes the m objective functions of the problem and $\mathbf{h}$ denotes the adaptation costs that would incur if transforming $\mathbf{u}_{t-1}$ to $\mathbf{u}_t$. When past decisions influence the evolution of $\mathbf{x}_t$ (e.g. if $\mathbf{B} \neq \mathbf{0}$ in Eq. (1)), then they also influence $\mathbf{f}$. On the other hand, even when the state transition is independent of past decisions, $\mathbf{f}$ may be not, e.g., when there are adaptation costs for changing previous decisions [25].

Definition 1 (Time Linkage (TL) regime): In a TL regime, either (a) the transition of $\mathbf{x}_t$ is influenced by previous stages decisions $\mathbf{u}_{t-k}$; and/or (b) the vector-valued performance $\mathbf{f}$ of $\mathbf{u}_t$ depends on $\mathbf{u}_{t-k}$.

We then denote the Stochastic PF (SPF) as $\mathcal{Z}_t^\star|\mathbf{u}_{t-1} = \mathcal{F}_t(\Omega_t^\star|\mathbf{u}_{t-1}, \mathbf{x}_t) = \{\mathbf{f}(\mathbf{u}_t^\star, \mathbf{x}_t, \mathbf{u}_{t-1}) : \mathbf{u}_t^\star \in \Omega_t^\star|\mathbf{u}_{t-1}\}$, where $\Omega_t^\star|\mathbf{u}_{t-1}$ is the PS at time t. It turns out that our SPF definition is based on the PD applied over the mean objective vectors:

Definition 2 (Stochastic PF, SPF): The SPF $\mathcal{Z}^\star = \mathcal{F}(\Omega^\star)$ is composed of all random objective vectors satisfying

$$(\forall \mathbf{z}^\star \in \mathcal{Z}^\star)\,(\nexists \mathbf{z}' \in \mathcal{F}(\Omega)) \text{ such that } \mathbb{E}[\mathbf{z}'] \preceq \mathbb{E}[\mathbf{z}^\star]. \quad (3)$$

The definition implies that the SPF is a random set whose support are subsets of the objective space $\mathcal{Z} = \mathcal{F}(\Omega)$.

Finally, chance constraints in the objective space are considered, for which no alternative is feasible with full certainty:

Definition 3 (ϵ-feasibility): Given a decision vector $\mathbf{u}_t$ with an associated random objective vector $\mathbf{z}_t = \mathbf{f}(\mathbf{u}_t, \mathbf{x}_t, \mathbf{u}_{t-1})$, we consider $\mathbf{u}_t$ to be ϵ-feasible if $\Pr\{\mathbf{z}_t \in \mathcal{Z}_t\} \geq \epsilon$.

B. Flexible Anticipatory Multi-Criteria Decision Making

The key to improve the range of future options is to anticipate flexible alternatives. Consider a finite objective set $\mathcal{Z}_t^N$ and let $\mathcal{Z}_{t+h}^{N\star}|\mathbf{u}_t^{\max}$ denote a future stochastic Pareto frontier approximation solving the AS-MOO model at time $t+h$, given that $\mathbf{u}_t^{\max} \in \mathcal{U}_t^N$ is chosen. The maximal future Hypv alternative $\mathbf{u}_t^{\max}$ satisfies

$$\mathbf{u}_t^{\max} = \arg\max_{\mathbf{u}_t \in \Omega_t} \mathbb{E}\left[\mathcal{S}\left(\sum_{h=1}^{H-1} \lambda_{t+h} \underbrace{\mathcal{Z}_{t+h}^{N\star}|\mathbf{u}_t}_{\text{future trade-offs}}\right)\right]. \quad (4)$$

From all $\mathbf{u}_t \in \Omega_t$ in Eq. (4), $\mathbf{u}_t^{\max}$ is foreseen to lead to future SPF approximations with maximal expected Hypv, where H is the anticipation horizon. The influence of future performance is mediated by anticipation rates, λ_{t+h}. Hypv is measured over a temporal convex combination between ordered objective distributions. Given two arbitrary sets, $\mathcal{A}$ and $\mathcal{B}$, the summation is defined for a lexical order over their elements, e.g., $\mathcal{A}+\mathcal{B} = \{a^{(1)}+b^{(1)}, \cdots, a^{(N)}+b^{(N)}\}$, so is the product $\lambda\mathcal{A} = \{\lambda a^{(1)}, \cdots, \lambda a^{(N)}\}$.

C. Model Formulation

The AS-MOO problem at time t depends on the solutions obtained for stages $t+1, \cdots, t+H-1$, and therefore follows a recurrence equation whose goal is to obtain the optimal anticipatory trajectory of SPF approximations such that

$$\mathcal{Z}_t^{N\star}|\mathbf{u}_{t-1}^{\max} = \arg\max_{\substack{\mathcal{Z}_t^N|\mathbf{u}_{t-1}^{\max} \\ \subset \mathcal{Z}_t|\mathbf{u}_{t-1}^{\max}}} \mathbb{E}\left[\mathcal{S}\left(\underbrace{\mathcal{Z}_t^N|\mathcal{Z}_{t+1:t+H-1}^{N\star}}_{\text{anticipatory trade-off sets}}\right)\right], \quad (5)$$

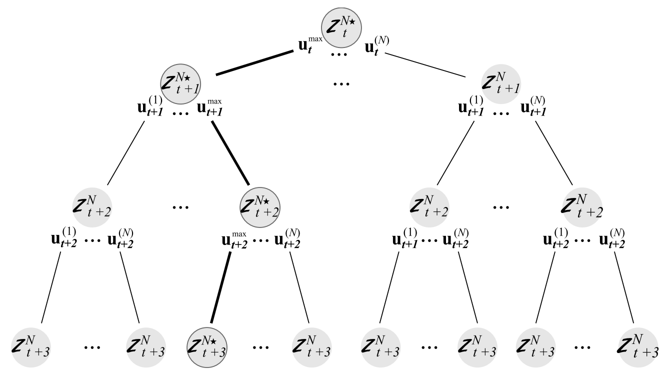


Fig. 3. In the AS-MOO methodology, a Pareto-efficient alternative $\mathbf{u}_t^{\max}$ is chosen among N options in $\mathcal{Z}_t^N$. This choice influences the future SPF approximations that can be formed to maximize the expected Hypv over discounted future efficient objective vectors, $\mathcal{Z}_{t+h}^{N\star}|\mathbf{u}_t^{\max}$.

where,

$$\underbrace{\mathcal{Z}_t^N|\mathcal{Z}_{t+1:t+H-1}^{N\star}}_{\text{anticipatory trade-offs}} \equiv \lambda_t \underbrace{\mathcal{Z}_t^N|\mathbf{u}_{t-1}}_{\text{current trade-offs}} + \sum_{h=1}^{H-1} \lambda_{t+h} \underbrace{\mathcal{Z}_{t+h}^{N\star}|\mathbf{u}_{t+h-1}^{\max}}_{\text{future trade-offs}}.$$

Note that any pair $\mathbf{z}_t^{(a)}, \mathbf{z}_t^{(b)} \in \mathcal{Z}_t^N$ is mutually non-dominated. The AS-MOO equation is also subject to the constraints:

$$\lambda_t + \sum_{h=1}^{H-1} \lambda_{t+h} = 1, \lambda_t \geq 0, \lambda_{t+h} \geq 0, \quad (6)$$

$$\Pr\{\mathbf{z}_{t+h} \in \mathcal{Z}_{t+h}\} \geq \epsilon, \ \forall\, h \in \{0, \cdots, H-1\}. \quad (7)$$

The model explicitly assesses the future discounted Pareto-efficient objective vectors, accumulating them with the current Pareto-efficient ones. The set $\mathcal{Z}_{t+h}^{N\star}|\mathbf{u}_t^{\max}$ represents the future SPF at time $t+h$ maximizing the expected $\mathcal{S}$ values computed for even further stages at $t+h+1, \cdots, t+h+H-1$. Thus, AS-MOO evaluates the consequences of implementing each $\mathbf{z}_t \in \mathcal{Z}_t$. It is worth noting that we assume the Markov property. The set $\mathcal{Z}_{t+h}^{N\star}|\mathbf{z}_t^{\max}$ then denotes the decision tree path (Fig. 3):

$$\mathcal{Z}_{t+1}^{N\star}|\mathbf{u}_t^{\max} \to \cdots \to \mathcal{Z}_{t+h}^{N\star}|\mathbf{u}_{t+h-1}^{\max} \equiv \mathcal{Z}_{t+h}^{N\star}|\mathbf{u}_t^{\max}. \quad (8)$$

Therefore, flexible $\mathbf{u}_{t+h}^{\max}$ choices must also be anticipated.

Note that the DM willingness for immediate performance – i.e., for maximal Hypv sets, given only current data – is maximal when $\lambda_t = 1$, what is equivalent to not looking ahead at all, whereas, for $\lambda_t = 0$, the DM is willing to achieve maximal Hypv in future stages, even if that means the set $\mathcal{Z}_t^{N\star}$ obtained by solving Eq. (5) does not yield the best possible performance when evaluated using only the current data.

D. Anticipating the Predicted Maximal Flexible Choice

In the AS-MOO solvers presented in Sec. IV-D, however, the future sets $\mathcal{U}_{t+h}^{N\star}|\mathbf{u}_t$ are replaced with the predicted sets $\hat{\mathcal{U}}_{t+1}^{N\star}|\mathbf{u}_t, \cdots, \hat{\mathcal{U}}_{t+H-1}^{N\star}|\mathbf{u}_t$ obtained from tracking the candidate trade-off solution vectors over time with Bayesian models. The future maximal Hypv choices are also replaced with the predictions $\hat{\mathbf{u}}_{t+1}^\star, \cdots, \hat{\mathbf{u}}_{t+H-1}^\star$, thus bypassing the recursive re-optimizations of the AS-MOO equations. Moreover, upon obtaining $\hat{\mathcal{U}}_t^\star$, the predicted maximal Hypv alternative $\hat{\mathbf{u}}_t^\star \in \hat{\mathcal{U}}_t^{N\star}$ is chosen from the trade-off set which was the solution to Eq. (5), which leads to the following definition.

Definition 4 (Anticipated Maximal Flexible Choice, AMFC): The AMFC $\hat{\mathbf{u}}_t^\star \in \hat{\mathcal{U}}_t^{N\star}$ is the one satisfying

$$\hat{\mathbf{u}}_t^\star = \underset{\hat{\mathbf{u}}_t \in \hat{\mathcal{U}}_t^{N\star}}{\arg\max}\, \mathbb{E}\left\{ \mathcal{S}\left(\sum_{h=1}^{H-1} \lambda_{t+h} \underbrace{\hat{\mathcal{Z}}_{t+h}^{N\star}|\hat{\mathbf{u}}_t}_{\text{predicted trade-offs}} \right) \right\}, \quad (9)$$

where $\hat{\mathcal{Z}}_{t+h}^{N\star}|\hat{\mathbf{u}}_t$ is the predicted maximal Hypv SPF approximation of N alternatives that would be obtained at time $t+h$, given that an alternative $\hat{\mathbf{u}}_t \in \hat{\mathcal{U}}_t^{N\star}$ is taken.

IV. Online Anticipatory Learning

While future Hypv maximization can postpone the assignment of relative importances over the decision criteria, the DM hesitation about time performance can be handled via Online Anticipatory Learning (OAL). OAL refer to methods for (i) self-adjusting the DM willingness to near-term performance; and (ii) for incorporating predictive knowledge into AS-MOO solvers, mediated by the perceived temporal predictability.

The assumption in OAL is that the more unpredictable the future, the more eager for immediate performance the DM should be, since, by construction, optimizing solutions for early performance is safer in this case. Conversely, the more certain the future, the more the DM is willing to capitalize on foreseen opportunities. OAL thus makes use of Bayesian tracking in both the objective and the search spaces to approximate AS-MOO solutions for which improved performance on the targeted environments is predicted, in terms of PD.

In AS-MOO, the set $\Lambda_{t:t+H-1} = \{\lambda_t, \cdots, \lambda_{t+H-1}\}$ encodes time preferences. Eliciting $\Lambda_{t:t+H-1}$ when N incomparable solutions are to be simultaneously searched for is challenging: the future outcomes of some alternatives may be more predictable than those of others. Hence, instead of pursuing a global set, we propose self-adjusting N independent sets, $\Lambda_{t:t+H-1}^{(1)}, \cdots, \Lambda_{t:t+H-1}^{(N)}$, so that portions of the SPF can be approximated under varying time preferences.

A. OAL in the Objective Space

The Kalman Filter (KF) [26] is a Bayesian tracking model providing closed-form expressions for updating posterior distributions if uncertainty can be modeled as a multivariate Gaussian and if the dynamics are linear. As a result, exact and computational inexpensive one-step ahead inference of the predictive distribution is available. The KF computes the best estimation (in the mean-squares sense) of a time-varying distribution under those assumptions. We thus model $\mathbf{z}_t \in \mathcal{Z}_t^N$ as a multivariate Gaussian over the $j = 1, \cdots, m$ decision criteria, i.e., $\mathbf{z}_t \sim \mathcal{N}(\mathbf{m}_{\mathbf{z}_t}, \mathbf{\Sigma}_{\mathbf{z}_t})$. The objective vectors in OAL are thus tracked using KF estimation (Fig. 4). The notation $\hat{\mathbf{z}}_{t+h}|\hat{\mathbf{z}}_t$ (for $h = 1, \cdots, H-1$) denotes h applications of the KF prediction step to yield long-term performance estimation:

$$\hat{\mathbf{z}}_{t+1}|\hat{\mathbf{z}}_t \to \cdots \to \hat{\mathbf{z}}_{t+h}|\hat{\mathbf{z}}_{t+h-1} \equiv \hat{\mathbf{z}}_{t+h}|\mathbf{z}_t. \quad (10)$$

We assume linear dynamics in the objective space: $\hat{\mathbf{z}}_t \sim \hat{\mathbf{z}}_{t-1} + \mathbf{B}\mathbf{u}_{t-1} + \dot{\mathbf{z}}_{t-1}$, i.e., the next state is distributed as the previous

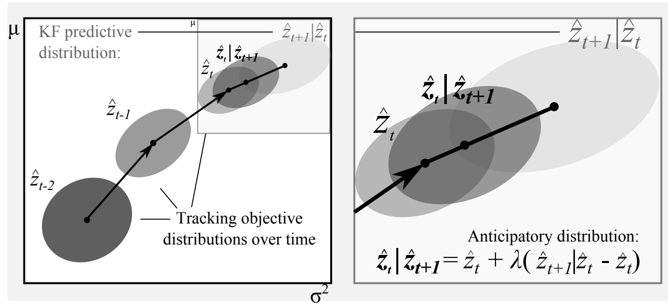


Fig. 4. On the left, the KF tracks performance for a fixed investment portfolio and estimates the predictive distribution $\hat{\mathbf{z}}_{t+1}|\hat{\mathbf{z}}_t$. The portfolio evolves from a low-risk/low-return estimative to a high-risk/high-return prediction. On the right, OAL is detailed. Given the up-to-date estimation $\hat{\mathbf{z}}_t$ and the predictive $\hat{\mathbf{z}}_{t+1}|\hat{\mathbf{z}}_t$, OAL produces an anticipatory distribution $\hat{\mathbf{z}}_t|\hat{\mathbf{z}}_{t+1}$ expressed as the convex combination between $\hat{\mathbf{z}}_t$ and $\hat{\mathbf{z}}_{t+1}|\hat{\mathbf{z}}_t$.

one plus a small displacement, that we call velocity, $\dot{\mathbf{z}}_t$. We thus track the extended state vector:

$$\mathbf{z}_t^+ = (z_{1,t} \cdots z_{m,t}\ \dot{z}_{1,t} \cdots \dot{z}_{m,t})^\intercal, \quad (11)$$

where m is the number of objective functions. It can be noted that the KF not only estimates the current objective values associated to a given $\mathbf{u}_t$, but also how such value is changing (i.e., by estimating the velocities). Those information are encoded in the joint distribution of $\mathbf{z}_t^+$ (Fig. 4).

1) Preference for Long-Term Predictability: The key for self-adjusting the $H \times N$ anticipation rates is to model a DM exhibiting preference for long-term predictability w.r.t. the PD over temporal performance. It is assumed that prediction is completely incorporated by a DM as one of the following probabilities approach one: the predicted objective distribution (a) dominates; (b) is dominated by; or (c) is incomparable to the current one. Prediction is thus ignored when the DM cannot tell above the chance level whether one of the three events (a)–(c) is satisfied, in which case near-term performance is preferred. Therefore, prediction is only useful to AS-MOO when it can yield safe information about the PD relation of temporal performance. In order to self-adjust the degree to which prediction is incorporated into AS-MOO, it suffices to estimate the probability for the incomparability case.

Definition 5 (Time Incomparability Probability, TIP): The probability of the current and the predicted objective vectors being mutually incomparable is expressed as

$$p_{t,t+h} = \Pr\left[\hat{\mathbf{z}}_t, \hat{\mathbf{z}}_{t+h}|\hat{\mathbf{z}}_t \text{ are mutually incomparable}\right]. \quad (12)$$

For two Gaussians, TIP can be queried from the marginal cumulative distributions. Because high TIPs convey high predictability regarding time incomparability, and considering the postponement of preference specification conveyed in the proposed flexible EMOO/MCDM methodology, we assume the DM prefers longer-term performance in this case.

Predictability is also high when TIP is low, in which case the DM can be confident that either (a) performance will improve so as to be preferable in the PD sense; or (b) performance will deteriorate and, thus, such alternative should be avoided. If, on the other hand, the estimated TIP is roughly $1/2$, unpredictability is maximal, which is indicative that the overlap between the temporal distributions is large. The anticipation rates can then be determined as:

$$\lambda_{t+h}^{(\mathcal{H})} = \frac{1}{H-1}\left[1 - \mathcal{H}(p_{t,t+h})\right] \tag{13}$$

where $\mathcal{H}$ is a binary entropy function, with $\mathcal{H}(1/2) = 1$ and $\mathcal{H}(0) = \mathcal{H}(1) = 0$, the parameter $p_{t,t+h}$ stands for the TIP value and the factor $\frac{1}{H-1}$ normalizes the sum terms in AS-MOO Eqs. (4) and (5) so that $\lambda_t + \sum_{h=1}^{H-1}\lambda_{t+h}^{(\mathcal{H})} = 1$.

Another measure of predictability considers the historical prediction error of the KF, in which case anticipation rates can be self-adjusted according to a normalized sum of KF squared residuals. The higher this quantity, the less confident the DM will be about the accuracy of the KF prediction, and vice-versa. Such retrospective assessment provides a means for self-adjusting the DM confidence and willingness to wait for future optimal performance in terms of how reliable past predictions were for a fixed $\mathbf{u}_t$.

2) The Anticipatory Objective Distribution: The self-adjustment of the anticipation rates is intended for controlling the extent to which prediction is accounted for guiding the search process toward flexible alternatives. This principle is captured in the following OAL definition[2]:

$$\hat{\mathbf{z}}_t|\hat{\mathbf{z}}_{t+1:t+H-1} \equiv \hat{\mathbf{z}}_t + \sum_{h=1}^{H-1}\lambda_{t+h}(\hat{\mathbf{z}}_{t+h}|\hat{\mathbf{z}}_t - \hat{\mathbf{z}}_t). \tag{14}$$

The equation states that the adjusted anticipatory distribution $\hat{\mathbf{z}}_t|\hat{\mathbf{z}}_{t+1:t+H-1}$ is a convex combination between the current KF estimation $\hat{\mathbf{z}}_t$ and the predictive $\hat{\mathbf{z}}_{t+h}|\hat{\mathbf{z}}_t$ distributions, where $\sum_{h=1}^{H-1}\lambda_{t+h}$ and $\lambda_t = 1 - \sum_{h=1}^{H-1}\lambda_{t+h}$. It turns out that the linear combination of two Gaussians, $\mathbf{x}$ and $\mathbf{y}$, through matrices $\mathbf{A}$ and $\mathbf{B}$ is given by:

$$\mathbf{Ax} + \mathbf{By} \sim \mathcal{N}(\mathbf{Am}_x + \mathbf{Bm}_y, \mathbf{A\Sigma}_x\mathbf{A}^\intercal + \mathbf{B\Sigma}_y\mathbf{B}^\intercal). \tag{15}$$

The resulting anticipatory distribution is thus (e.g. for $T = 2$):

$$\begin{aligned}\hat{\mathbf{z}}_t|\hat{\mathbf{z}}_{t+1} \sim \mathcal{N}(&\lambda_t\mathbf{m}_{\mathbf{z}_t} + \lambda_{t+1}\mathbf{m}_{\hat{\mathbf{z}}_{t+1}|\mathbf{z}_t},\\ &\lambda_t^2\mathbf{\Sigma}_{\mathbf{z}_t} + \lambda_{t+1}^2\mathbf{\Sigma}_{\hat{\mathbf{z}}_{t+1}|\mathbf{z}_t}),\end{aligned} \tag{16}$$

where $\lambda_t = 1 - \lambda_{t+1}$. OAL (Fig. 4) is thus expressed in the Gaussian parameters. The covariance $\mathbf{\Sigma}$ is required to be positive semidefinite and, in fact, any nonnegative linear combination of such matrices is positive semidefinite [27].

B. OAL in the Search Space

The goal of OAL in the search space is to learn to which directions each Pareto-efficient decision is moving so as to predict what they will be like in future stages. For the PO application, the search space is the $(d-1)$-simplex $\Omega_t = S^{d-1} \subset \mathcal{R}^d = \left\{\mathbf{u} \in \mathcal{R}^d : \sum_{k=1}^{d} u_k = 1, u_k \geq 0 \ \forall\ k\right\}$, where d is the number of search variables. Let $\mathbf{u}_t \in \Omega_t$ be a random vector lying in S^{d-1}. A sound choice for representing uncertainty that takes S^{d-1} as support is the Dirichlet Distribution (DD) [28]. A DD vector is denoted as $\mathbf{u} \sim \mathcal{D}(\boldsymbol{\alpha})$, where $\boldsymbol{\alpha} = (\alpha_1 \ \cdots \ \alpha_d)^\intercal$ determines the shape of the DD (with $\alpha_k > 0$ for all k). The DD can be parameterized by the mean $\mathbf{m} = \left(\frac{\alpha_1}{\alpha_C} \cdots \frac{\alpha_d}{\alpha_C}\right)^\intercal$, where $\alpha_C = \sum_k \alpha_k$ is the concentration controlling dispersion around $\mathbf{m}$.

[2] The notation $\mathbf{z}_{t+1:t+H-1}$ denotes a sequence $\mathbf{z}_{t+1}, \cdots, \mathbf{z}_{t+H-1}$.

1) A Sliding Window Dirichlet Dynamical Model: In Eq. (5), the sets $\mathcal{U}_{t+h}^{N\star}|\mathbf{u}_t^{\max}$ must be computed in advance. While exponential $O(N^H)$ offline re-optimizations are required to anticipate $\mathbf{u}_t^{\max}$ (see Fig. 3), we propose instead predicting N unconditional DDs in $\hat{\mathcal{U}}_{t+h}^{N\star}$ for alleviating the need to exactly compute $\mathbf{u}_t^{\max}$. Although prediction provides a less accurate estimative, the computational cost to obtain it online is negligible (bilinear complexity, $O(HN)$) when compared to recursively re-optimizing. The expected value $\mathbb{E}\left[\hat{\mathcal{U}}_{t+h}^{N\star}\right]$ is then used as a surrogate to $\mathcal{U}_{t+h}^{N\star}|\mathbf{u}_t^{\max}$ for approximately solving the online sequence of AS-MOO problems.

We propose updating the posterior distributions by removing portfolios from the DD process history, so uncertainty can either decrease or increase over time. A window size K then limits the collection of portfolios previous to $t-K$. Let $\boldsymbol{\alpha}_t^{(i)}|\mathbf{u}_{t-1}^{(i)}$ denote the update of concentration $\boldsymbol{\alpha}_t^{(i)}$ given that evidence on $\mathbf{u}_{t-1}^{(i)}$ was collected and let the posterior be

$$\begin{aligned}&\hat{\mathbf{u}}_0^{(i)} \sim \mathcal{D}(\boldsymbol{\alpha}_0^{(i)}), \hat{\mathbf{u}}_1^{(i)}|\hat{\mathbf{u}}_0^{(i)} \sim \mathcal{D}(\boldsymbol{\alpha}_1^{(i)}|\hat{\mathbf{u}}_0^{(i)}),\\ &\qquad \cdots, \hat{\mathbf{u}}_t^{(i)}|\mathbf{u}_{t-1}^{(i)} \sim \mathcal{D}(\boldsymbol{\alpha}_t^{(i)}|\mathbf{u}_{t-1}^{(i)}).\end{aligned} \tag{17}$$

Assuming uniform concentration at $t = 0$, we have the following recursive update:

$$\boldsymbol{\alpha}_t^{(i)}|\mathbf{u}_{t-1}^{(i)} = \begin{cases} \boldsymbol{\alpha}_{t-1}^{(i)}|\mathbf{u}_{t-2}^{(i)} + s\mathbf{u}_{t-1}^{(i)}, & \text{if } t < K,\\ \boldsymbol{\alpha}_{t-1}^{(i)}|\mathbf{u}_{t-2}^{(i)} + s\mathbf{u}_{t-1}^{(i)} - \boldsymbol{\alpha}_0^{(i)}, & \text{if } t = K,\\ \boldsymbol{\alpha}_{t-1}^{(i)}|\mathbf{u}_{t-2}^{(i)} + s\mathbf{u}_{t-1}^{(i)} - s\mathbf{u}_{t-K-1}^{(i)}, & \text{otherwise.}\end{cases} \tag{18}$$

We denote $s\mathbf{u}_{t-1}^{(i)} - s\mathbf{u}_{t-K-1}^{(i)}$ as the velocity of the DDs in the latest K stages. The velocities encode the underlying dynamical model. Also, for $K = 1$, note how $\hat{\mathbf{u}}_t^{(i)}$ depends only on the previous portfolios, $s\mathbf{u}_{t-1}^{(i)}$. The scale s controls the dispersion of the DD and must be set in advance, whereas the concentration of $\hat{\mathbf{u}}_t^{(i)}|\mathbf{u}_{t-1}^{(i)}$ goes through a K-fold increase from $t = 0$ to $t = K$ and remains constant for $t > K$, as historical portfolios give place to more recent ones.

Thus, the posterior uncertainty in Eq. (18) decreases after K stages and remains constant for $t \geq K$. The estimation of $\mathbb{E}\left[\hat{\mathcal{U}}_{t+h}^{N\star}\right]$ is performed independently for each $\hat{\mathbf{u}}_t \in \hat{\mathcal{U}}_{t+h}^{N\star}$. In order to map each $\hat{\mathbf{u}}_t \in \hat{\mathcal{U}}_t^{N\star}$ to each $\hat{\mathbf{u}}_{t+h} \in \hat{\mathcal{U}}_{t+h}^{N\star}$, we sort all decisions w.r.t. to a lexical order. Let $\hat{\mathbf{u}}_{t+h}^{(i)}$ be the i-th rank decision w.r.t. to the first criterion. For example, let $\hat{\mathcal{U}}_t^{N\star} = \{\hat{\mathbf{u}}_{t,1}, \hat{\mathbf{u}}_{t,2}, \hat{\mathbf{u}}_{t,3}\}$ and $\hat{\mathbf{m}}_{\mathbf{z}_{t,1}} = (1\ 1)^\intercal, \hat{\mathbf{m}}_{\mathbf{z}_{t,2}} = (0\ 2)^\intercal, \hat{\mathbf{m}}_{\mathbf{z}_{t,3}} = (2\ 0)^\intercal$. When sorting $\hat{\mathcal{U}}_t^{N\star}$ in increasing order, we obtain $\hat{\mathbf{u}}_t^{(1)} = \hat{\mathbf{u}}_{t,2}, \hat{\mathbf{u}}_t^{(2)} = \hat{\mathbf{u}}_{t,1}, \hat{\mathbf{u}}_t^{(3)} = \hat{\mathbf{u}}_{t,3}$.

When predicting the mean vectors $\hat{\mathbf{m}}_{\hat{\mathbf{u}}_{t+h}^{(i)}}|\mathbf{m}_{\mathbf{u}_{t-1}^{(i)}}$, for $h = 1, \cdots, T$, we propose a linear dynamical model with constant velocity, i.e., $\dot{\mathbf{m}}_{\hat{\mathbf{u}}_t^{(i)}} = \dot{\mathbf{m}}_{\mathbf{u}_{t-1}^{(i)}}$. Thus, in the case of $K = 1$, defining the velocities for the DD mean vectors as $\dot{\mathbf{m}}_{\hat{\mathbf{u}}_t^{(i)}} = \mathbf{m}_{\hat{\mathbf{u}}_t^{(i)}} - \mathbf{m}_{\mathbf{u}_{t-1}^{(i)}}$, and assuming one-step ahead prediction, $\hat{\mathbf{m}}_{\hat{\mathbf{u}}_{t+1}^{(i)}} = \mathbf{m}_{\mathbf{u}_{t-1}^{(i)}} + 2\dot{\mathbf{m}}_{\mathbf{u}_{t-1}^{(i)}}$.

Input: The anticipation horizon H
Input: The concentration scaling parameter s
Input: A candidate $\hat{\mathbf{u}}_t \in \hat{\mathcal{U}}_t$ and the $\hat{\mathcal{U}}_t$ set
Input: The sets $\hat{\mathcal{U}}_{t-K:t}^N$ and $\mathcal{X}_{t-K:t-1} = \{\mathbf{x}_{t-K}, \cdots, \mathbf{x}_{t-1}\}$
Output: The estimated anticipatory distribution for $\hat{\mathbf{u}}_t$

procedure ANTICIPATORYDISTRIBUTION
  Compute the rank i of $\hat{\mathbf{u}}_t$ w.r.t. $\mathcal{U}_t$, yielding $\hat{\mathbf{u}}_t^{(i)}$
  Set $\hat{\mathcal{Z}}_{t:t+H-1}^{(i)} \leftarrow \emptyset$
  Retrieve rank i decisions in $\hat{\mathcal{U}}_{t-K:t}^{(i)} \subset \hat{\mathcal{U}}_{t-K:t}$
  for $h = 0$ **to** $H-1$ **do**
    Predict $\hat{\mathbf{u}}_{t+h}^{(i)}$ using $\hat{\mathcal{U}}_{t-K:t}^{(i)}$ with Eqs. 19–22
    Predict $\hat{\mathbf{z}}_{t+h}^{(i)}$ using $\hat{\mathcal{Z}}_{t-K:t}^{(i)}$ with the KF (Fig. 4)
    $\hat{\mathbf{z}}_{t+h}^{(i)} \leftarrow \hat{\mathbf{z}}_{t+h}^{(i)} + \mathbf{h}(\mathbf{m}_{\hat{\mathbf{u}}_{t+h}^{(i)}}, \mathbf{m}_{\hat{\mathbf{u}}_{t+h-1}^{(i)}})$
    $\hat{\mathcal{Z}}_{t:t+H-1}^{(i)} \leftarrow \hat{\mathcal{Z}}_{t:t+H-1}^{(i)} \cup \{\hat{\mathbf{z}}_{t+h}^{(i)}\}$
  end for
  Apply OAL over $\hat{\mathcal{Z}}_{t:t+H-1}^{(i)}$ using Eq. (14)
  return The anticipatory distribution $\hat{\mathbf{z}}_t^{(i)} | \hat{\mathbf{z}}_{t+1:t+H-1}^{(i)}$
end procedure

Pseudocode 1: Anticipatory Distribution Estimation

We use $\upsilon_{t+1} \in \left[\frac{1}{2}, 1\right]$ for adjusting the belief that the vector motion will keep the same velocity as measured at the preceding stage. The confidence is maximal for $\upsilon_{t+1} = 1$, whereas for $\upsilon_{t+1} = \frac{1}{2}$, it is assumed the decision vector will not change, being equivalent to the static model $\hat{\mathbf{m}}_{\hat{\mathbf{u}}_{t+1}^{(i)}} = \mathbf{m}_{\hat{\mathbf{u}}_t^{(i)}}$:

$$\hat{\mathbf{m}}_{\hat{\mathbf{u}}_{t+1}^{(i)}} = \mathbf{m}_{\mathbf{u}_{t-1}^{(i)}} + \upsilon_{t+1} 2\dot{\mathbf{m}}_{\mathbf{u}_{t-1}^{(i)}}. \tag{19}$$

We then self-adjust υ_{t+1} as the complement of the binary entropy measured for the TIP between $\mathbf{u}_t^{(i)}$ and $\mathbf{u}_{t-1}^{(i)}$, for which case we use the KF estimation $\hat{\mathbf{z}}_t^{(i)} | \mathbf{z}_{t-1}^{(i)}$ and $\mathbf{z}_{t-1}^{(i)}$:

$$\upsilon_{t+1} = 1 - \frac{1}{2}\mathcal{H}(p_{t-1,t}), \tag{20}$$

where $p_{t-1,t}$ is the TIP between $\hat{\mathbf{z}}_t^{(i)} | \mathbf{z}_{t-1}^{(i)}$ and $\mathbf{z}_{t-1}^{(i)}$. With this strategy, we propagate the uncertainty about whether the current i-th ranked decision vector $\hat{\mathbf{u}}_t^{(i)}$ at t is incomparable with respect to the preceding i-th decision at $t-1$, $\mathbf{u}_{t-1}^{(i)}$. Therefore, our surrogate estimate for $\mathcal{U}_{t+1}^{N\star}$ is obtained as:

$$\mathbb{E}\left[\hat{\mathcal{U}}_{t+h}^{N\star}\right] \approx \left\{\hat{\mathbf{m}}_{\hat{\mathbf{u}}_{t+h}^{(1)\star}}, \cdots, \hat{\mathbf{m}}_{\hat{\mathbf{u}}_{t+h}^{(N)\star}}\right\}, \tag{21}$$

and the recursive update step for correcting $\hat{m}_{\hat{u}_{l,t+h}^{(i)\star}}$ upon the observation of the d candidate mean decision variables $m_{\hat{u}_{l,t+h}^{(i)}}$ at $t+h$ (for $l = 1, \cdots, d$) is then as follows [29]:

$$\hat{m}_{\hat{u}_{l,t+h}^{(i)\star}} | \hat{u}_{l,t+h}^{(i)} = \hat{m}_{\hat{u}_{l,t+h}^{(i)\star}} + \mathrm{Var}\left[\hat{u}_{l,t+h}^{(i)\star}\right] \frac{m_{\hat{u}_{l,t+h}^{(i)}} - \hat{m}_{\hat{u}_{l,t+h}^{(i)\star}}}{\hat{m}_{\hat{u}_{l,t+h}^{(i)\star}}(1 - \hat{m}_{\hat{u}_{l,t+h}^{(i)\star}})}. \tag{22}$$

Pseudocode 1 describes the estimation of the anticipatory distribution $\hat{\mathbf{z}}_t^{(i)} | \hat{\mathbf{z}}_{t+1:t+H-1}^{(i)}$ for each $\hat{\mathbf{u}}_t^{(i)} \in \hat{\mathcal{U}}_t^\star$. Tracking and prediction begin in the search space, after which the KF is applied for the resulting objective vectors, which are then adjusted with the predicted cost component $\mathbf{h}$ (see Eq. (2)).

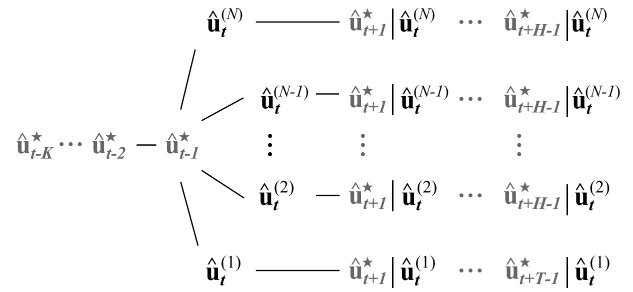


Fig. 5. The N possible predictive trajectories for the sequence of future $(H-1) \times N$ AMFCs over time, conditioned on each available candidate Pareto-efficient decision in the estimated essential set $\hat{\mathcal{U}}_t^{N\star}$.

C. Anticipating the Predicted Maximal Flexible Choice

In order to anticipate the AMFC $\hat{\mathbf{u}}_t^\star \in \hat{\mathcal{U}}_t^{N\star}$, tracking and prediction is performed for (a) each decision in $\hat{\mathcal{U}}_{t+h}^{N\star}$; (b) the corresponding objective vectors $\hat{\mathbf{z}}_{t+h}^{(i)}$; and (c) the sequence of implemented AMFCs $\hat{\mathbf{u}}_{t+h}^\star$ (see Fig. 5):

Step 1 Predict the sets $\hat{\mathcal{U}}_{t+1}^N, \cdots, \hat{\mathcal{U}}_{t+H-1}^N$ element-wise;
Step 2 Predict the component $\mathbf{g}$ (Eq. (2)): $\{\hat{\mathbf{z}}_{t+h}^{(i)}\}_{h=1}^{H-1}$;
Step 3 Predict $(H-1) \times N$ AMFCs (see Fig. 5) conditioned at $\hat{\mathbf{u}}_t^{(i)} \in \hat{\mathcal{U}}_t^{N\star}$: $\left\{\hat{\mathbf{u}}_{t+h}^\star | \hat{\mathbf{u}}_t^{(i)}\right\}_{h=1}^{H-1}$;
Step 4 Predict the N^2 cost-adjusted objective distributions, given $\hat{\mathbf{u}}_t^{(j)}$ $(i, j \in \{1, \cdots, N\})$: $\{\hat{\mathbf{z}}_{t+h}^{(i)} | \hat{\mathbf{u}}_t^{(j)}\}_{h=1}^{H-1}$;
Step 5 Apply OAL and identify the AMFC using E Monte Carlo simulated scenarios (see Sec. V-D):

$$\hat{\mathbf{u}}_t^\star = \arg \max_{\hat{\mathbf{u}}_t \in \hat{\mathcal{U}}_t^{N\star}} \frac{1}{E} \sum_{e=1}^{E} \mathcal{S}\left(\sum_{h=1}^{H-1} \lambda_{t+h} \left\{\hat{\mathbf{z}}_{e,t+h}^{(i)} | \hat{\mathbf{u}}_t\right\}_{i=1}^{N}\right). \tag{23}$$

D. The Anticipatory SMS-EMOA (ASMS-EMOA)

In the $\mathcal{S}$-Metric Selection algorithm (SMS-EMOA) [30], decisions yielding higher Hypv contribution ($\Delta_\mathcal{S}$) to their class $\mathcal{C}$ are preferred, for which $\Delta_\mathcal{S}(\mathbf{z}, \mathcal{C}) = \mathcal{S}(\mathcal{C}) - \mathcal{S}(\mathcal{C} \setminus \{\mathbf{z}\})$. By iteratively replacing the least Hypv contributing solutions, the hope in SMS-EMOA is that Hypv can be maximized. We thus propose the Anticipatory SMS (ASMS)[3] for approximately solving AS-MOO, using Bayesian tracking, implementing OAL, and computing $\mathbb{E}[\Delta_\mathcal{S}]$ over the estimated anticipatory distributions. The Pseudocodes 2 and 3 describe the ASMS algorithm, whose outputs are (1) the set of mutually non-dominated (w.r.t. the mean vectors) anticipatory distributions, and the AMFC ($\hat{\mathbf{u}}_t^\star \in \hat{\mathcal{U}}_t^{N\star}$) satisfying Eq. (23).

E. Computing Expected Anticipatory Hypv Contributions

Let $\hat{\mathbf{z}}_t^{(i)} | \hat{\mathbf{z}}_{t+h}^{(i)} \in \mathcal{C}$ ($|\mathcal{C}| \geq 3$) be a non-extremal bivariate[4] anticipatory distribution. Let $\mathbf{g} = (g_1\ g_2)^\intercal$ be the objective functions. Assuming e.g. min g_1 and max g_2, and given two neighboring $\hat{\mathbf{z}}_t^{(i-1)}, \hat{\mathbf{z}}_t^{(i+1)} \in \mathcal{C}$, we express $\Delta_\mathcal{S}(\hat{\mathbf{z}}_t^{(i)})$ in terms

[3] Source code available at http://dx.doi.org/10.6084/m9.figshare.1330398
[4] We write $\hat{\mathbf{z}}_t^{(i)} \equiv \hat{\mathbf{z}}_t^{(i)} | \hat{\mathbf{z}}_{t+h}^{(i)}$ and $\hat{z}_{j,t}^{(i)} \equiv \hat{z}_{j,t}^{(i)} | \hat{z}_{j,t+h}^{(i)}$ for simplicity.

Input: The anticipation horizon H
Input: $\hat{\mathcal{U}}^N_{t-K:t-1}$ and the implemented AMFCs $\left\{\mathbf{u}^\star_{t-h}\right\}^K_{h=1}$
Input: Historical $\mathcal{X}_{t-K:t-1} = \{\mathbf{x}_{t-K}, \cdots, \mathbf{x}_{t-1}\}$
Output: A set $\hat{\mathcal{U}}^{N\star}_t$ approximately satisfying Eq. (5)
Output: A AMFC $\hat{\mathbf{u}}^\star_t$ satisfying Eq. (9)
procedure ASMS-EMOA
  Set the generation count as $g \leftarrow 0$
  Set the current population as $\hat{\mathcal{U}}^N_t \leftarrow \hat{\mathcal{U}}^N_{t-1}$
  for $i = 1$ **to** N **do**
    Compute $\underbrace{\hat{\mathbf{z}}^{(i)}_t|\hat{\mathbf{z}}^{(i)}_{t+1:t+H-1}}_{\text{anticipatory distribution}}$ ▷ Pseudocode 1
  end for
  repeat
    Compute $\{\mathcal{C}_j\}^C_{j=1}$ from $\mathbb{E}\left[\{\hat{\mathbf{z}}^{(i)}_t|\hat{\mathbf{z}}^{(i)}_{t+1:t+H-1}\}^N_{i=1}\right]$
    Compute $\mathbb{E}\left[\Delta_{\mathcal{S}}(\hat{\mathbf{z}}^{(i)}_t|\hat{\mathbf{z}}^{(i)}_{t+1:t+H-1}, \mathcal{C}_j)\right]$, for each $\hat{\mathbf{z}}^{(i)}_t|\hat{\mathbf{z}}^{(i)}_{t+1:t+H-1} \in \mathcal{C}_j$ and $1 \leq j \leq C$
    Generate a new decision $\hat{\mathbf{u}}'_t$ from $\hat{\mathcal{U}}^N_t$
    Augment $\hat{\mathcal{U}}^{N+1}_t \leftarrow \hat{\mathcal{U}}^N_t \cup \{\hat{\mathbf{u}}'_t\}$
    Compute $\hat{\mathbf{z}}'_t|\mathbf{z}'_{t+1:t+H-1}$ ▷ Pseudocode 1
    Reduce $\hat{\mathcal{U}}^N_t \leftarrow$ REDUCE($\hat{\mathcal{U}}^{N+1}_t$)
    Advance to the next generation $g \leftarrow g + 1$
  until Stopping criteria are not met
  Compute $\hat{\mathbf{u}}^\star_t$ using Eq. (23) ▷ Section IV-C
  return $\left\{\hat{\mathcal{U}}^{N\star}_t, \hat{\mathbf{u}}^\star_t\right\}$
end procedure

Pseudocode 2: The Anticipatory $\mathcal{S}$-Metric Selection EMOA

Input: An augmented candidate set $\hat{\mathcal{U}}^{N+1}_t$
Output: A reduced candidate set $\hat{\mathcal{U}}^N_t$
procedure REDUCE
  Compute $\{\mathcal{C}_j\}^C_{j=1}$ from $\mathbb{E}\left[\{\hat{\mathbf{z}}^{(i)}_t|\hat{\mathbf{z}}^{(i)}_{t+1:t+H-1}\}^{N+1}_{i=1}\right]$
  Select $\hat{\mathbf{u}}^* = \arg\min_{\hat{\mathbf{u}}\in\mathcal{C}_C} \mathbb{E}\left[\Delta_{\mathcal{S}}(\hat{\mathbf{u}}, \mathcal{C}_C)\right]$
  return $\hat{\mathcal{U}}^{N+1}_t \setminus \{\hat{\mathbf{u}}^*\}$
end procedure

Pseudocode 3: Reduce $\mathcal{U}^{N+1}_t$

of the j-th objective given that the k-th objective (for $k, j \in \{1, 2\}$ and $k \neq j$) is fixed at its mean (m_k) (see Fig. 6):

$$\Delta_{\mathcal{S}}(\hat{\mathbf{z}}^{(i)}_t) = (\hat{z}^{(i)}_{1,t}|\hat{m}^{(i)}_{2,t} - \hat{z}^{(i-1)}_{1,t}|\hat{m}^{(i-1)}_{2,t}) \times (\hat{z}^{(i+1)}_{2,t}|\hat{m}^{(i+1)}_{1,t} - \hat{z}^{(i)}_{2,t}|\hat{m}^{(i)}_{1,t}). \quad (24)$$

In Eq. (24), we make use of conditionals, which are also Gaussians, and we need to estimate the expected $\Delta_{\mathcal{S}}(\hat{\mathbf{z}}^{(i)}_t)$.

Theorem 1: The expected value of $\Delta_{\mathcal{S}}(\hat{\mathbf{z}}^{(i)}_t)$ is

$$\begin{aligned}\mathbb{E}\left[\Delta_{\mathcal{S}}(\hat{\mathbf{z}}^{(i)}_t)\right] &= \left(\hat{m}^{(i)}_{1,t} - \hat{m}^{(i-1)}_{1,t}\right) \times \left(\hat{m}^{(i+1)}_{2,t} - \hat{m}^{(i)}_{2,t}\right) \\ &+ \mathrm{Cov}(\hat{z}^{(i)}_{1,t}|\hat{m}^{(i)}_{2,t},\ \ \hat{z}^{(i+1)}_{2,t}|\hat{m}^{(i+1)}_{1,t}) \\ &- \mathrm{Cov}(\hat{z}^{(i)}_{1,t}|\hat{m}^{(i)}_{2,t},\ \ \hat{z}^{(i)}_{2,t}|\hat{m}^{(i)}_{1,t}) \\ &- \mathrm{Cov}(\hat{z}^{(i-1)}_{1,t}|\hat{m}^{(i-1)}_{2,t},\ \ \hat{z}^{(i+1)}_{2,t}|\hat{m}^{(i+1)}_{1,t}) \\ &+ \mathrm{Cov}(\hat{z}^{(i-1)}_{1,t}|\hat{m}^{(i-1)}_{2,t},\ \ \hat{z}^{(i)}_{2,t}|\hat{m}^{(i)}_{1,t}). \quad (25)\end{aligned}$$

Proof: See Azevedo [31]. ■

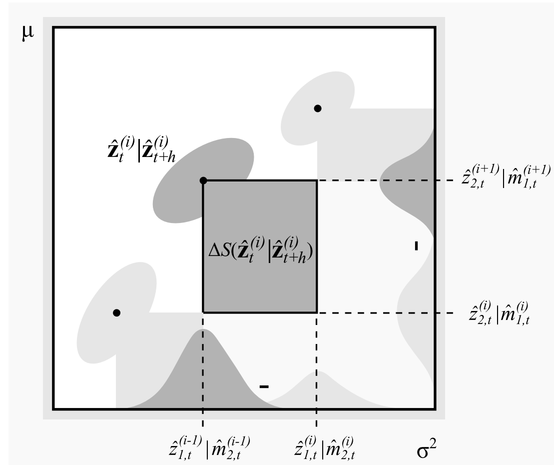


Fig. 6. Computing $\mathbb{E}\left[\Delta_{\mathcal{S}}\right]$ in the objective space by conditioning the decision criteria at their mean values, assuming max return and min risk.

Note $\mathbb{E}\left[\Delta_{\mathcal{S}}(\hat{\mathbf{z}}^{(i)}_t|\hat{\mathbf{z}}^{(i)}_{t+h})\right]$ depends not only on the mean vectors of $\hat{\mathbf{z}}^{(i)}_t|\hat{\mathbf{z}}^{(i)}_{t+h}$ and of the class neighbors ($\hat{\mathbf{z}}^{(i-1)}_t|\hat{\mathbf{z}}^{(i-1)}_{t+h}$ and $\hat{\mathbf{z}}^{(i+1)}_t|\hat{\mathbf{z}}^{(i+1)}_{t+h}$), but also on the covariances of the conditional anticipatory distributions. Therefore, the learned covariances provide uncertainty awareness in AS-MOO.

V. Experimental Design

The aim of the experimental design is to investigate the effects on the observed sample average Hypv computed over *future out-of-sample investment periods* of the following two factors: (1.a) tracking and prediction in the decision and objective spaces to evolve an anticipatory Stochastic Pareto Frontier (SPF), compared to (1.b) the assumption of a static SPF of a myopic optimizer; and (2.a) identifying and implementing Anticipated Maximal Flexible Choices (AMFC) over time, compared to (2.b) selecting the portfolio corresponding to the median point of the evolved SPF; and to (2.c) randomly selecting portfolios from the evolved Pareto set. We name those the (1) anticipation and the (2) DM factors.

A. Investigated Algorithmic Variants

The anticipation factor is controlled by four levels of window size (K): $K \in \{0, 1, 2, 3\}$. For $K = 0$, we have the myopic baseline SMS [30] which assumes the environment to be static, i.e., $\hat{\mathbf{u}}_{t+h} \sim \hat{\mathbf{u}}_t$ and $\hat{\mathbf{z}}_{t+h} \sim \hat{\mathbf{z}}_t$. That assumption implies the estimated TIPs (Eq. (12)) equal $1/2$ and, hence, all anticipation rates are self-adjusted to $\lambda^{(i)}_t = 1$ ($i = 1, \cdots, N$), i.e., immediate outcomes are given full weight. For $K > 0$, the past solutions and objective distributions obtained in the latest K periods serve as input to the DD and KF tracking methods, in which case anticipation is enabled.

The DM factor determines whether (a) AMFCs; (b) the median portfolio from the evolved SPF; or (c) randomly drawn portfolios from the Pareto Set are implemented. The goal is to contrast a DM that aims to explicitly maximize future freedom of choice (Hypv DM, HDM) from a DM that aims to select an alternative providing a balanced trade-off (median DM, MDM) and from an indifferent DM (Random DM, RDM). Therefore, six algorithmic variants are investigated: (1.a) + (2.a) = ASMS/HDM; (1.a) + (2.b) = ASMS/MDM; (1.a) + (2.c) = ASMS/RDM; (1.b) + (2.a) = SMS/HDM; (1.b) + (2.b) = SMS/MDM; and (1.b) + (2.c) = SMS/RDM.

B. AS-MOO Application in Portfolio Optimization

Let $\mathbf{r} \in \mathcal{R}^N$ be the return vector of N risky assets, in which $\boldsymbol{\mu}_\mathbf{r}$ and $\boldsymbol{\Sigma}_\mathbf{r}$ are its mean and covariance, and let $\mathbf{u} \in S^{N-1}$ denote the proportions of wealth to be invested. Then, the *classical Markowitz PO formulation* for risk minimization [32] is: $\mathbf{u}^\star = \arg\min_\mathbf{u} \{\mathbf{u}^\intercal \boldsymbol{\Sigma}_\mathbf{r} \mathbf{u} : \boldsymbol{\mu}_\mathbf{r}^\intercal \mathbf{u} \geq r_0\}$, in which r_0 is the minimum acceptable return. The application of AS-MOO for approximating portfolios lying in the SPF can reveal the trade-offs between expected return maximization and expected risk minimization within a nonstationary and probabilistic investment environment. In AS-MOO, the investment decision leading to future Pareto Sets with maximal expected Hypv is chosen as a basis for rebalancing the current implemented portfolio. The state parameters $\mathcal{X}_t = \left\{\hat{\boldsymbol{\mu}}_{\mathbf{r},t}, \hat{\boldsymbol{\Sigma}}_{\mathbf{r},t}\right\}$ must be estimated from the available data. We then propose the sequential formulation

$$\min_{\mathbf{u}_t} \quad \mathbf{u}_t^\intercal \hat{\boldsymbol{\Sigma}}_{\mathbf{r},t} \mathbf{u}_t \tag{26}$$

$$\max_{\mathbf{u}_t} \quad \hat{\boldsymbol{\mu}}_{\mathbf{r},t}^\intercal \mathbf{u}_t - h(\mathbf{u}_t, \mathbf{u}_{t-1}^\star) \tag{27}$$

$$\text{s.t.} \quad c_l \leq c(\mathbf{u}_t) \leq c_u, \tag{28}$$

where $c(\mathbf{u}_t)$ computes the number of assets in $\mathbf{u}_t$ with non-zero weight ($u_t > 0$). Following the AS-MOO notation, $\mathbf{z}_t | \mathbf{u}_{t-1}^\star = \mathbf{g}(\mathbf{u}_t, \mathcal{X}_t) + \mathbf{h}(\mathbf{u}_t, \mathbf{u}_{t-1}^\star)$, where:

$$\mathbf{g}(\mathbf{u}_t, \mathcal{X}_t) = \left(\mathbf{u}_t^\intercal \hat{\boldsymbol{\Sigma}}_{\mathbf{r},t} \mathbf{u}_t \quad \hat{\boldsymbol{\mu}}_{\mathbf{r},t}^\intercal \mathbf{u}_t\right)^\intercal, \text{ and} \tag{29}$$

$$\mathbf{h}(\mathbf{u}_t, \mathbf{u}_{t-1}^\star) = \left(0 \quad h(\mathbf{u}_t, \mathbf{u}_{t-1}^\star)\right)^\intercal, \tag{30}$$

in which h is a cost function representing all incurring transaction fees. The risk of $\mathbf{u}_t$ is not to be affected by the previous portfolio.

C. Simulated Benchmark Data Generator

In order to assess the behavior of the AS-MOO methodology under controlled conditions, we devised a random dynamic PO instance generator, with two factors: (1) the periodicity; and (2) the severity with which disruptive changes affect the joint return distribution. The modeled disruption is an additive noise process, in which the parameters are perturbed as $\mathcal{X}_\text{new} = (1-\eta)\mathcal{X}_\text{old} + \eta \mathcal{P}_\text{random}$, where $\mathcal{P}_\text{random}$ is a random set of parameters and η controls the severity with which noise is absorbed. Let $\mathcal{X}_1 = \{\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1\}$ be the initial set of parameters, where $\boldsymbol{\mu}_1$ is the mean vector and $\boldsymbol{\Sigma}_1$ is the covariance matrix. Let t_0 be the period when the first disruptive change occurs. The set of parameters at period t_0 is generated as $\mathcal{X}_{t_0} = (1-\eta)\mathcal{X}_1 + \eta \mathcal{P}_{t_0}$. Parameters at $t \in (1, t_0)$ undergo smooth linear dynamics:

$$\mathcal{X}_t = \left(1 - \frac{t}{t_0}\right)\mathcal{X}_1 + \frac{t}{t_0}\mathcal{X}_{t_0}. \tag{31}$$

In the general case, the k-th disruptive event will yield:

$$\mathcal{X}_{t_{k-1}} = (1-\eta)\mathcal{X}_{t_{k-1}} + \eta \mathcal{P}_{t_k}. \tag{32}$$

Any distribution over $\boldsymbol{\mu}_{t_k}$ can be used for generating $\mathcal{P}_{t_k}$. In this paper, we use a uniform distribution over the hyperbox $[\mu_\text{lb}, \mu_\text{ub}]^N$, where μ_lb and μ_ub are the lower and upper bounds of return for all assets. For generating random covariance matrices, we use a Gram Matrix-based method [33], which consists of uniformly sampling unit vectors in the unit hypersphere $\mathcal{G}^{N-1}$, i.e., $\frac{\mathbf{x}}{||\mathbf{x}||} \sim U(\mathcal{G}^{N-1})$, with $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and taking $\rho_\mathcal{P} = \mathbf{x}\mathbf{x}^\intercal$. The matrix is then built as $\boldsymbol{\Sigma}_\mathcal{P}(i,j) = \{\rho_\mathcal{P}(i,j)\sigma_i\sigma_j\}$, where σ_i is the standard deviation of the i-th asset, uniformly sampled from $[\sigma_\text{lb}, \sigma_\text{ub}]$.

D. Experimental Methodology

Let $\mathcal{W}_t$ be the initial wealth to be invested in a probabilistic market and let set $t \leftarrow K$. Given a stream of T known Gaussian returns parameters $\{\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t\}_{t=1}^T$ (T is the number of stages), the following procedures are repeated in all ASMS variants at each stage:

Step 1 Sample a total of $S \times K$ return vectors from each available $\mathcal{N}\left(\boldsymbol{\mu}_{t-K}, \boldsymbol{\Sigma}_{t-K}\right), \cdots, \mathcal{N}\left(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t\right)$, where E is the sample size;

Step 2 Estimate $\left\{\hat{\mathcal{X}}_{t-k}\right\}_{k=1}^K = \left\{\hat{\boldsymbol{\mu}}_{t-k}, \hat{\boldsymbol{\Sigma}}_{t-k}\right\}_{k=1}^K$ from the set of simulated returns;

Step 3 Use the parameter stream $\left\{\hat{\mathcal{X}}_{t-k}\right\}_{k=1}^K$ for solving AS-MOO at the t-th stage to obtain $\hat{\mathcal{U}}_t^{N\star}$ and $\hat{\mathbf{u}}_t^\star$;

Step 4 Use the out-of-sample future test state parameters $\mathcal{X}_{t+1}$ to compute $\hat{\mathcal{Z}}_{t+1} = \mathbf{f}\left(\hat{U}_t^{N\star}, \mathcal{X}_{t+1}, \hat{\mathbf{m}}_{\mathbf{u}_t^\star}\right)$ and collect the estimated sample average future Hypv:

$$\hat{\mathcal{S}}_{t+1} = \frac{1}{E}\sum_{e=1}^{E} \mathcal{S}\left(\hat{\mathcal{Z}}_{e,t+1}, \mathbf{z}_\text{ref}\right); \tag{33}$$

Step 5 Implement the selected mean portfolio $\hat{\mathbf{m}}_{\mathbf{u}_t^\star}$ from the evolved SPF, advance to $t \leftarrow t+1$, and update

$$\begin{aligned} \mathcal{W}_t \leftarrow \mathcal{W}_{t-1} + &\text{ mean observed return for } \hat{\mathbf{u}}_t^\star \\ - &\text{ transaction costs of transforming} \\ &\hat{\mathbf{m}}_{\mathbf{u}_{t-1}^\star} \text{ into } \hat{\mathbf{m}}_{\mathbf{u}_t^\star}. \end{aligned} \tag{34}$$

We set the initial wealth to $\mathcal{W}_0 = 10,000.00$ monetary units, and construct the cost function $\mathbf{h}$ following a suggestion by the Securities Commission of Brazil which is widespread, practiced by the major brokers operating in the São Paulo BOVESPA stock market. The brokerage costs depend on the value of trade, combining fixed and proportional fees (Tab. I).

In order to assess the statistical significance of the anticipation and DM factors, we use two-way ANOVA. Pairwise significance is assessed with the non-parametric Mann–Whitney U test. All algorithms are evaluated from 30 independent runs.

1) Prediction Of Change in Direction, POCID: The POCID captures the proportion of correct predictions w.r.t. the directions toward which the risk and expected return of a fixed

TABLE I
BROKERAGE FEES UTILIZED IN THE BRAZILIAN STOCK MARKET.

Traded value	Proportional Fee	Fixed Fee
< 135.07	0.0%	2.70
≥ 135.08 and < 498.62	2.0%	0.00
≥ 498.63 and $< 1,514.69$	1.5%	2.49
$\geq 1,514.70$ and $< 3,029.38$	1.0%	10.06
$\geq 3,029.39$	0.5%	25.21

portfolio evolve between subsequent periods:

$$\text{POCID}(j) = \frac{1}{N \times (T-1)} \sum_{i=1}^{N} \sum_{t=1}^{T-1} D(j,i,t), \tag{35}$$

$$D(j,i,t) = \begin{cases} 1, & \text{if } (z_{j,t+1}^{(i)\star} - z_{j,t}^{(i)\star})(\hat{m}_{\hat{z}_{j,t+1}^{(i)\star}} - \hat{m}_{\hat{z}_{j,t}^{(i)\star}}) \geq 0, \\ 0, & \text{otherwise,} \end{cases}$$

where T is the number of decision stages. We consider POCID = $\frac{1}{2} \times$ (POCID(return) + POCID(risk)). In Eq. (35), $z_{j,t+1}^{(i)\star}$ and $z_{j,t}^{(i)\star}$ are two objective values observed for the i-th ranked portfolio in subsequent periods w.r.t. the j-th objective function, whereas $\hat{m}_{\hat{z}_{j,t+1}^{(i)\star}}$ and $\hat{m}_{\hat{z}_{j,t}^{(i)\star}}$ are the corresponding predicted mean values obtained via KF tracking.

E. Parameters Settings and Datasets

We now describe the data collection and experimental setup.

1) Artificial and Real-World Datasets: A total of 10 investment scenarios are assessed: (a) seven artificial; and (b) three real-world stock markets, including London FTSE 100, Dow Jones Index (DJI), and Hong Kong's Hang Seng Index (HSI). All benchmarks provide $A = 30$ simulated assets for composing the portfolios, whereas for the real-world instances we have $A = 87$ for FTSE; $A = 30$ for DJI; and $A = 49$ for HSI. The total number of periods for all instances is $T = 25$[5]. The real-world scenarios are composed of daily adjusted close prices collected between $11/20/2006-12/31/2012$, from which 50 days lagged returns were computed: $r_{n,k} = \frac{\mathcal{V}_{n,k+50} - \mathcal{V}_{n,k}}{\mathcal{V}_{n,k+50}}$, where $\mathcal{V}_{n,k+50}$ is the value of the n-th asset at day $k + 50$. Each stage comprises a period of one and a half year of data. Between each stage, the 50 oldest lagged returns are discarded and the 50 latest ones are included in the sample from which the parameters $\left\{\hat{\boldsymbol{\mu}}_t, \hat{\boldsymbol{\Sigma}}_t\right\}_{t=1}^{T}$ are estimated, where 50 business days roughly corresponds to two and a half months. Thus, the stage $t = 1$ in FTSE comprises data from $11/20/2006 - 05/20/2008$, $t = 2$ to $02/01/2007 - 07/30/2008$, and so forth.

As for the artificial benchmarks, we investigate severities of $\eta \in \{0.5, 1.0\}$ and periodicities of $\tau \in \{2, 4, 8\}$. We refer to the six instances as PO(η,τ). The seventh instance was generated to simulate an environment in which the covariance $\boldsymbol{\Sigma}_t$ varies at every four periods, but that is static only w.r.t. the mean returns $\boldsymbol{\mu}_t$. We refer to it as the "static mean PO".

[5]Except for FTSE with $T = 24$ due to data availability issues at the time the experiments were executed.

2) Bayesian Tracking Parameters: Tracking with KF and the DD MAP requires setting (a) the covariance matrices representing the initial uncertainty and the measurement noise, for the KF; and (b) the scale factor s representing the initial dispersion, for the DD tracking. In the former case, the initial covariance was set directly from the data, using the sample covariances for the objective distributions evaluated at $t = 0$, whereas the noise covariance was set to $\frac{1}{E} \times \Sigma_t$, what corresponds to the covariances of the sample means. In all experiments, $E = 1000$, i.e., the risk and return is evaluated from 1000 simulated daily return values. The scale factor in DD tracking is set to $s = 1$, after finding that its influence is negligible in preliminary tests. The anticipation horizon utilized was $H = 2$ (one-step-ahead prediction).

3) Constraint Handling: We considered minimum and maximum cardinality of $c_l = 5$ and $c_u = 15$ assets. Moreover, the feasible objective space includes all portfolios attaining mean return of at least 0% and at most 20% risk. The feasibility probability was set to 99% ($\epsilon = 0.99$, see Def. 3). Constraint handling in ASMS is done as in [34].

4) ASMS Parameters: Population size in all variants was set to $N = 20$ and 30 generations were executed for each stage. Seeding was utilized so that the previous portfolios serve as the new starting points in subsequent periods. Mutation rate was set to 0.3; crossover probability to 0.2; and binary tournament selection was utilized over the sample mean vectors, with expected Hypv contribution (Eq. (24)) as a tiebreaker. The reference point for computing Hypv was set to $\mathbf{z}^{\text{ref}} = (0.2, 0.0)^\mathsf{T}$ in terms of risk and return, coinciding with the objectives feasibility boundaries. Finally, we used the same crossover and mutation operators as in Azevedo and Von Zuben [24].

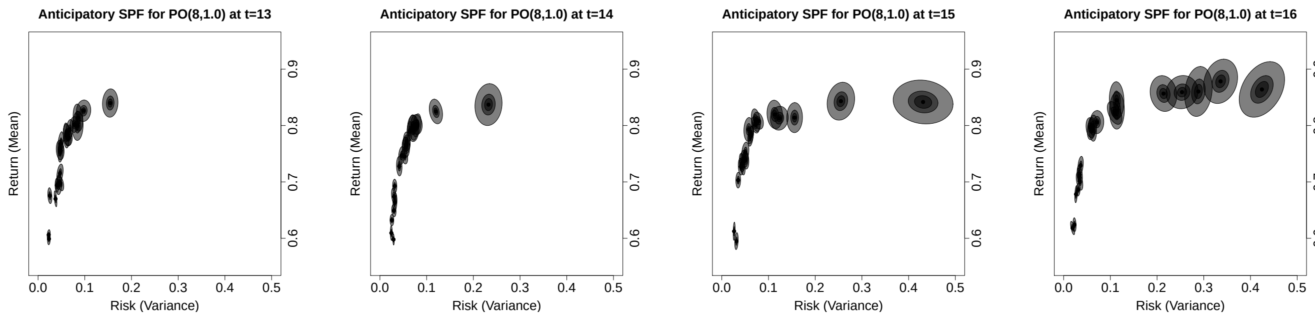
VI. RESULTS AND DISCUSSION

All experiments consider AS-MOO instances for which trade-off solutions are optimized according to historical data and one of them is deployed in a time-varying, post-optimization unseen environment. The implemented solution affects the objective function evaluations of subsequent periods, thus modifying the admissible future Pareto sets.

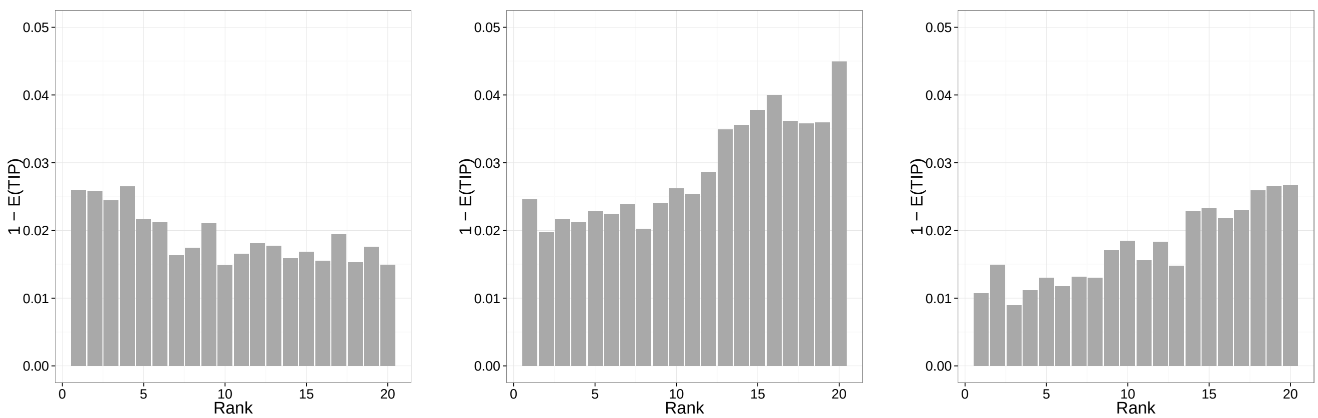
A. Estimated Confidence Over the Approximated SPFs

The differences in the estimated anticipatory distributions are shown in Fig. 7 (a)–(d), for 4 evolved SPFs in one run of ASMS/HDM at 4 arbitrarily chosen periods of the PO(8,1.0) instance. Note that high return portfolios are associated to high risk. Moreover, the variances of the anticipatory distributions are larger for portfolios attaining high risk/return when compared to the low risk/return ones. This result is consistent with the intuition that risky portfolios tend to be less stable over time [31]. The KF tracking seems to have correctly captured this uncertainty.

The barplots shown in Fig. 7 (e)–(g) for the DJI instance depict the predictability degrees (confidence) for the evolved portfolios, expressed in terms of Eq. (13). For a small window size of $K = 1$, the confidences were higher for portfolios achieving higher returns, whereas for larger window sizes ($K = 2, 3$), this tendency was reversed. Different yet orderly



(a) SPF approximation at t=13 (b) SPF approximation at t=14 (c) SPF approximation at t=15 (d) SPF approximation at t=16



(e) Confidence (K=1) (f) Confidence (K=2) (g) Confidence (K=3)

Fig. 7. Example of anticipatory SPFs evolved with ASMS/HDM in (a)–(d). The estimated confidence (complement of entropy) distributions over the evolved SPFs, averaged over all stages, are shown in parts (e)–(g) for DJI. Rank 1 corresponds to highest return, whereas rank 20 corresponds to the lowest return.

confidence distributions were observed for the remaining instances. The emergence of such tendencies suggests that OAL effectively self-adapted the anticipation rates, according to the market dynamics and the chosen window size.

B. Bayesian Tracking Influence on POCID

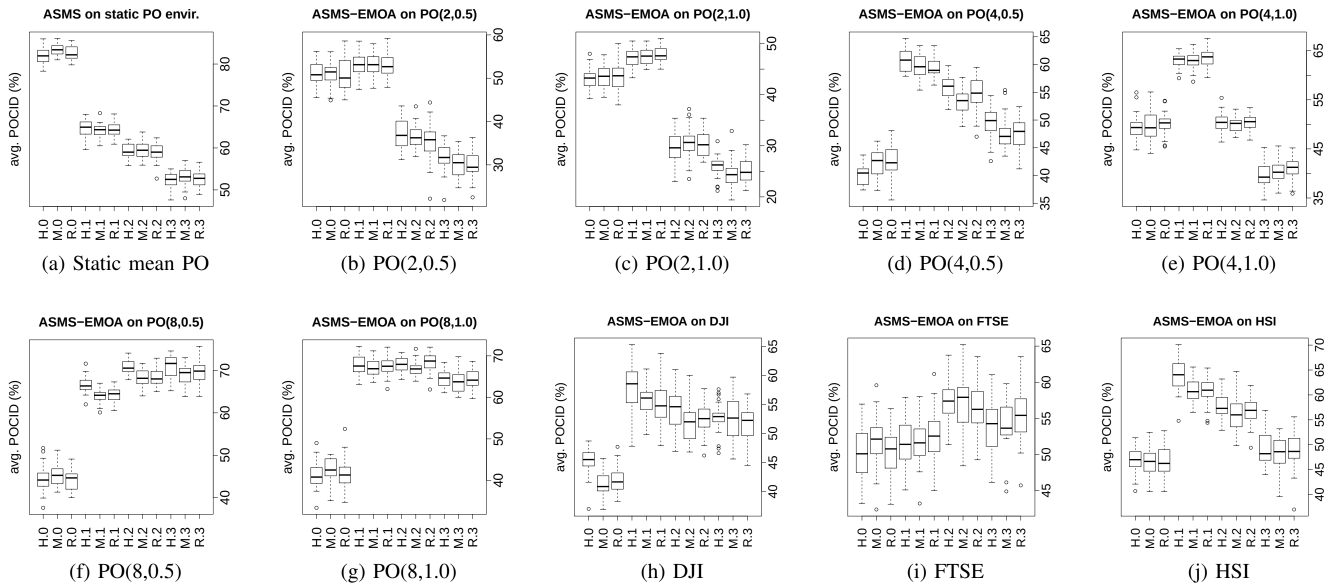
The effectiveness of prediction in ASMS is analyzed in terms of POCID (Eq. (35)). The results for the PO benchmarks are shown in Fig. 8 (a)–(g). As expected for the static mean returns scenario (see Sec. V-E1), the best median POCID results (roughly 85%) were obtained with stationary assumption ($K = 0$, see Fig. 8 (a)). It is likely that, for $K > 0$, tracking may have been severely affected by the sampling noise. As for the remaining instances, prediction was more effective when disruptive changes occurred at low frequency (every $\tau = 8$ periods), with moderate severity ($\eta = 0.5$). The highest median POCID in this case ($\approx 72\%$) was observed with the largest window size, $K = 3$ (Fig. 8 (f)), whereas, for the faster-paced environments ($\tau = 2, 4$), the shortest window size ($K = 1$) allowed for the best performance. The results for the real-world datasets are shown in Fig. 8 (h)–(j), for which tracking was significantly more effective with $K = 2$, for FTSE, and with $K = 1$ for DJI and HSI, which corroborates the intuition that fast changing distributions require responsive tracking capable of quickly discarding uncorrelated past data.

In addition, the stationary assumption (for $K = 0$) was significantly outperformed in terms of POCID for at least one value of $K > 0$ in all 9 nonstationary instances, including the real-world markets, which suggests that OAL has succeeded on learning portfolios attaining good predictability, as captured by POCID. Another noteworthy effect in POCID was that of the DM factor for PO(4,0.5), PO(8,0.5), and DJI, which suggests that rebalancing the implemented portfolios according to the HDM strategy may lead to more predictable Pareto sets. Finally, it is worth mentioning the similarity of the POCID patterns obtained for the real-world DJI and HSI datasets to those observed for the artificial PO(4,0.5), which suggests that the changing dynamics of those markets may have been roughly approximated with our benchmark generator.

C. Effects on the Out-Of-Sample Future Average Hypervolume

We assess whether the proposed methodology has succeeded in attaining its main goal: handling undefined preferences by improving future freedom of choice. The figure of merit is the out-of-sample future average Hypv ($\hat{\mathcal{S}}_{t+1}$, see Eq. (33)) computed from the obtained Pareto sets at each stage. Table II shows the two-way ANOVA results for the anticipation (WS) and DM effects on $\hat{\mathcal{S}}_{t+1}$, besides their interaction described in the "WS:DM" rows. The DM factor had a significant effect on future Hypv in all seven artificial scenarios (85%).

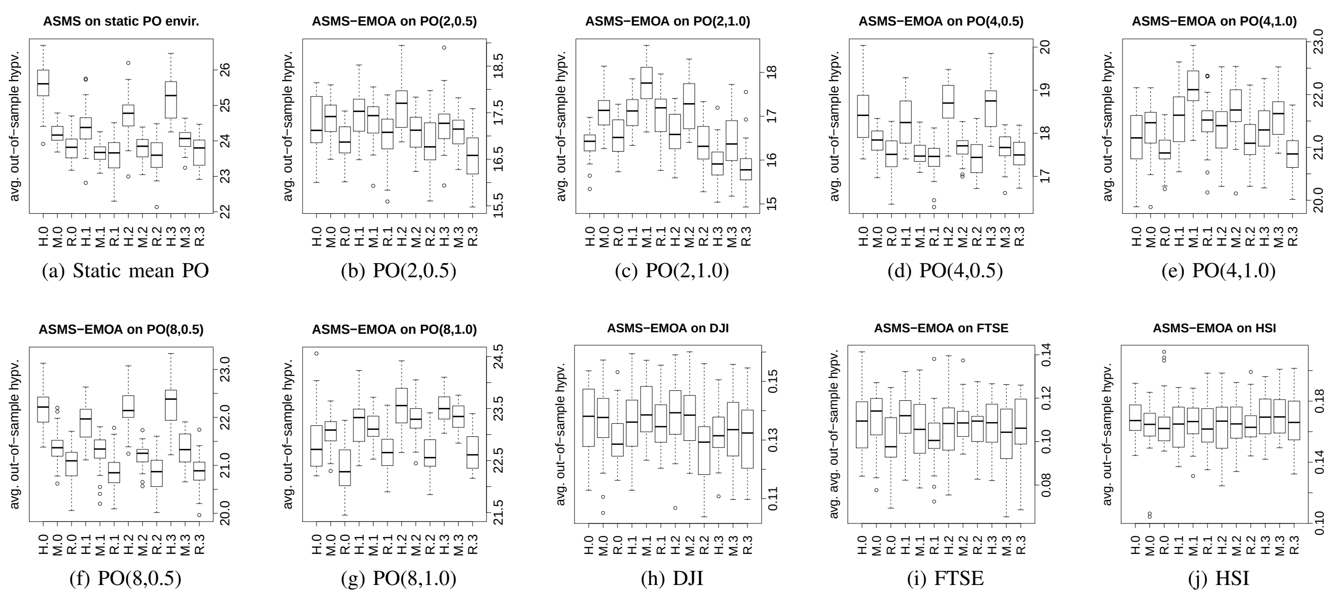
Inspecting Fig. 9, the superiority of the HDM over the RDM becomes clear, for the artificial scenarios, whereas MDM outperformed HDM and RDM on the most challenging artificial instances, PO(2,1.0) and PO(4,1.0), in terms of future



(a) Static mean PO (b) PO(2,0.5) (c) PO(2,1.0) (d) PO(4,0.5) (e) PO(4,1.0)

(f) PO(8,0.5) (g) PO(8,1.0) (h) DJI (i) FTSE (j) HSI

Fig. 8. Boxplots of Prediction Of Change in Direction (POCID). H stands for HDM, M for MDM, and R for RDM within the notation (DM.K).



(a) Static mean PO (b) PO(2,0.5) (c) PO(2,1.0) (d) PO(4,0.5) (e) PO(4,1.0)

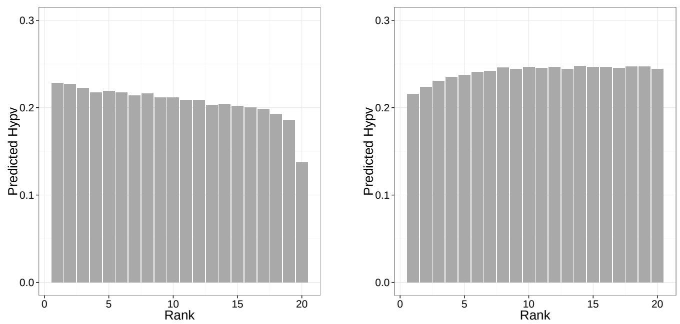
(f) PO(8,0.5) (g) PO(8,1.0) (h) DJI (i) FTSE (j) HSI

Fig. 9. Boxplots of Out-Of-Sample Future Hypv. H stands for HDM, M for MDM, and R for RDM within the notation (DM.K).

out-of-sample average Hypv. Those instances correspond to the highest disruptive environments, wherein the joint return distribution parameters change in a completely random way, with maximum severity, after each two and four consecutive periods of time, respectively. This makes it very hard to accurately perform Bayesian tracking. This result corroborates the intuition that, in the impossibility of obtaining accurate prediction, the median DM strategy would provide a good heuristic for retaining future flexibility. On the remaining instances, for which there is more predictability, the HDM obtained a significant edge on most scenarios, with MDM also outperforming RDM with statistical significance in most cases.

Moreover, from the real-world ANOVA results, the DM factor was significant on the FTSE dataset. Besides, the results in Fig. 9 reveal that the HDM also achieved significantly higher $\hat{\mathcal{S}}_{t+1}$ values than those achieved by the RDM on 4 out of 12 real-world cases, considering FTSE, DJI, and HSI and the four window size levels, being statistically equivalent to RDM for the remaining cases. As for the anticipation factor, significant effects on future Hypv were detected for five out of seven artificial scenarios (71%). The exceptions were PO(8,0.5) and PO(4,0.5), although the effects would have been significant at the 10% level.

Also, ANOVA revealed significant interaction between the anticipation and DM factors in static mean PO, PO(2,0.5), and PO(8,1.0), where the effects of window size depended on the DM factor: for both HDM and MDM, varying K had significant effects on $\hat{\mathcal{S}}_{t+1}$, whereas the same cannot be said



(a) Predicted Hypv (K=0) in DJI (b) Predicted Hypv (K=3) in DJI

Fig. 10. Predicted future expected Hypv in ASMS/HDM for the obtained ordered Pareto sets, averaged over all stages and experiments.

TABLE II
PEARSON CORRELATION COEFFICIENTS (** $p < 0.01$; * $p < 0.05$)

Instance	Coherence vs. $\hat{\mathcal{S}}_{t+1}$	Cardinality vs. $\hat{\mathcal{S}}_{t+1}$
Static	0.69**	0.21**
PO(2,0.5)	0.42**	-0.07*
PO(2,1.0)	0.20**	-0.02
PO(4,0.5)	0.26**	-0.30**
PO(4,1.0)	0.39**	-0.04
PO(8,0.5)	0.14**	-0.04
PO(8,1.0)	0.29**	-0.04
DJI	-0.09*	0.11**
FTSE	0.20**	-0.10**
HSI	-0.26**	0.10**

TABLE III
ANOVA FOR WS AND DM FACTORS ON $\hat{\mathcal{S}}_{t+1}$ (** $p < 0.01$; * $p < 0.05$).

AS-MOO Instance	Factor	Mean Sq	F value	Significance
Static	WS	189.80	21.22	**
	DM	1496.00	167.24	**
	WS:DM	36.50	4.08	**
PO(2, 0.5)	WS	43.89	6.66	**
	DM	260.20	39.50	**
	WS:DM	18.19	2.76	*
PO(2, 1.0)	WS	554.3	49.400	**
	DM	422.60	37.66	**
	WS:DM	6.80	0.60	≈
PO(4, 0.5)	WS	21.40	2.40	*
	DM	919.5	103.17	**
	WS:DM	10.60	1.19	≈
PO(4, 1.0)	WS	145.80	24.154	**
	DM	229.15	37.963	**
	WS:DM	5.81	0.963	≈
PO(8, 0.5)	WS	18.10	2.47	≈
	DM	1190.20	162.47	**
	WS:DM	7.80	1.07	≈
PO(8, 1.0)	WS	86.90	24.07	**
	DM	457.30	126.60	**
	WS:DM	12.90	3.57	**
DJI	WS	0.01	0.48	≈
	DM	0.03	1.86	≈
	WS:DM	0.01	0.36	≈
FTSE	WS	0.00	0.47	≈
	DM	0.02	4.92	**
	WS:DM	0.01	1.65	≈
HSI	WS	0.01	0.47	≈
	DM	0.00	0.06	≈
	WS:DM	0.00	0.15	≈

for RDM. In the remaining artificial instances (see Fig. 9), varying K has lead to significant effects on $\hat{\mathcal{S}}_{t+1}$ independent of the DM type, with the exception of PO(8,0.5).

Figure 10 shows the average predicted Hypv associated to the evolved trade-off portfolios, from which the patterns of choice of the HDM variant can be inferred. Remarkably, the tendency observed for the confidence distribution in Fig. 7 (e)–(g) is present for the predicted Hypv on DJI: when $K = 0$, the portfolios predicted to yield maximal future Hypv tended to be associated to higher return/risk, whereas the trend was reversed for $K > 1$. Thus, the myopic SMS/HDM ($K = 0$) tended to produce more risky choices, whereas ASMS/HDM ($K > 0$) tended to choose flexible, less risky portfolios. This behavioral shift may be associated to the improved predictability of the less risky portfolios (comparing $K > 0$ to $K = 0$), as captured by the confidence patterns in Fig. 7. We regard the association between higher predicted future Hypv and higher estimated predictability as evidence that the ASMS/HDM variant may indeed favor safer and flexible decision trajectories, as intended in OAL (see Fig. 2).

Finally, there has been at least one value of $K > 0$ for which ASMS/HDM has outperformed the baseline SMS-EMOA/RDM in terms of $\hat{\mathcal{S}}_{t+1}$, for the whole test suite. Considering all 30 settings from the combination of $K \in \{1, 2, 3\}$ with the 10 instances, ASMS/HDM significantly outperformed SMS-EMOA/RDM in 26 out of the 30 (87%) cases. When the comparison is between HDM and MDM, this statistic is 12 out of 30 (40%). Thus, irrespective of the dependency of ASMS on the tracking parameters (i.e., K), the significantly greater $\hat{\mathcal{S}}_{t+1}$ values achieved by ASMS/HDM are indicative that the AS-MOO methodology has attained the research main goal.

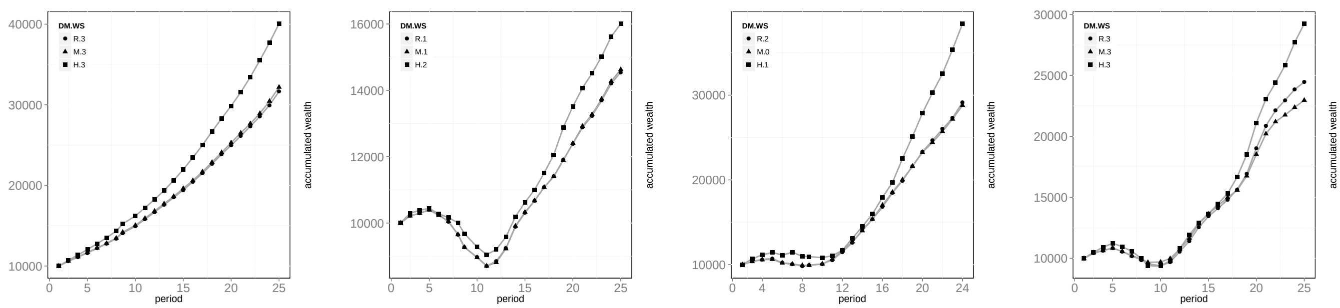
D. Correlation between $\hat{\mathcal{S}}_{t+1}$ and the Portfolio Compositions

In order to assess the portfolio compositions, we propose a coherence measure for the Pareto sets. It consists of summing over the cosine similarities of each portfolio to the population centroid, $\mathbf{u}_t^C$. The Pearson correlation of $\hat{\mathcal{S}}_{t+1}$ with (1) coherence; and (2) the cardinality of the portfolios (see Tab. III) was computed over all settings ($K \in \{0, 1, 2, 3\}$ and $\mathrm{DM} \in \{\mathrm{HDM}, \mathrm{MDM}, \mathrm{RDM}\}$). Coherence was significantly positively correlated with $\hat{\mathcal{S}}_{t+1}$ for 8 out of the 10 instances. On the other hand, correlation between cardinality and $\hat{\mathcal{S}}_{t+1}$ was significantly negative for 3 out of the 10 instances.

Those results indicate that the DM factor may induce ASMS to favor coherent populations so to mitigate the costs arising from the DM choices on future Hypv. On the other hand, the significant negative correlation coefficients observed for DJI and HSI may indicate that, in some scenarios, more population diversity – i.e., lower coherence – may be required in order improve future freedom of choice. The correlation patterns for cardinality vs. $\hat{\mathcal{S}}_{t+1}$ were not as significant, but we suspect that lower cardinality may be generally associated to reduced adaptation costs so to improve $\hat{\mathcal{S}}_{t+1}$.

E. Dynamics of the Investor Wealth

An example of the average expected wealth of an investor implementing the choices produced by ASMS variants for PO(8,1.0) is given in Fig. 11 (a). It can be noted the HDM has outperformed both the MDM and the RDM (this was true



(a) PO(8,1.0) (b) DJI (c) FTSE (d) HSI

Fig. 11. Average wealth accumulated throughout all investment periods, by rebalancing the implemented portfolio according to the HDM, MDM, and RDM variants. H stands for HDM, M for MDM, and R for RDM within the notation (DM.K). The chosen K values correspond to the best results for each strategy.

for every $K > 0$). The results for the real-world datasets are given in Fig. 11 (b)–(d), for which similar patterns of wealth evolution were observed. It can be noted that all three modeled investors have faced negative returns between periods 5–11, in all three instances, although the HDM had a significant edge in the FTSE and DJI markets.

Interestingly, the period $t = 11$ corresponds to the starting date of 03/08/2009, the day after the FTSE 100 index closed at a historical six-year 3,700 points lower mark, during the financial crisis. That day later proved to be the tuning point for a steady market uphill recovery that still endures to this day. For the HSI, the RDM strategy outperformed MDM, whereas the HDM always outperformed both MDM and RDM. Those results suggest that, although expected wealth was not directly optimized for, anticipating flexible decisions with the proposed ASMS/HDM system may not necessarily conflict with wealth maximization and can actually help the DM on achieving higher returns. This result is consistent with the account of Benjaafar et al. [1] discussed in Sec. II.

VII. Concluding Remarks

This paper proposed the online maximization of a sound set-based quality indicator in Multi-Objective Optimization (MOO) – the Hypervolume (Hypv) – for handling dynamic, stochastic, and cost-adjusted objective functions. It was then argued that implementing solutions predicted to maximize the expected Hypv of future Pareto sets can be used as a strategy for maximizing future freedom of choice and for robustly postponing preference specification. The reasoning is that, if preferences cannot be determined a priori in such uncertain environments, the Decision Maker (DM) would be better off implementing flexible solutions (i) predicted to keep his/her set of future options open; and (ii) whose future consequences attain higher degrees of predictability.

Online Anticipatory Learning (OAL) was then designed to self-adjust time preferences in response to the perceived temporal Pareto dominance predictability. Because there are no proposals in Evolutionary MOO (EMOO) meeting the problem requirements, an existing Hypv-based EMOO algorithm (SMS-EMOA [30]) was adapted to our methodology. Moreover, a portfolio optimization random instance generator was designed for providing controllable experimental scenarios.

The results for both simulated and real-world markets showed that the anticipatory methodology was competent on obtaining sets of portfolios yielding significantly higher average Hypv in most instances, when evaluated for out-of-sample future data, compared to baseline strategies. Therefore, we conclude that the proposed system has attained the goal of improving future freedom of choice. On the other hand, for environments exhibiting very low levels of predictability, we acknowledge that the proposed Bayesian tracking system might not be able to properly identify the correct maximal flexible choices and, thus, we recommend the DM to focus on trade-off alternatives ordered around the median point of the estimated Stochastic Pareto Frontier, in the bi-objective case.

The successful widespread application of Anticipatory Stochastic MOO (AS-MOO) models and algorithms will require competent system identification and proper modeling assumptions that may capture the dynamics of the problem under consideration. On the other hand, this provides several research opportunities for improving the anticipatory EMOO models and algorithms proposed in this paper.

A. Future Works

Alternative predictive models should be investigated for taking advantage of relationships between the composition decision vectors and future Hypv. For instance, it has been suggested that historical diversity levels might provide good predictive information for potential of Hypv improvement in the approximation dynamics of metaheuristics [35].

One next step for research on AS-MOO is to integrate OAL into other Hypv-based EMOO algorithms, such as IBEA [36]. Nevertheless, practical challenges must be addressed, such as the derivation of the expected value of IBEA's fitness equation, under simple assumptions.

Finally, the design of new AS-MOO benchmark problems and instances must be pursued in order to investigate whether future Hypv maximization may lead to a general problem-solving strategy, as realized in Wissner-Gross and Freer [15] with future causal entropy maximization.

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Carlos R. B. Azevedo received his Dr.E.E. degree from the University of Campinas (Unicamp), Brazil, in 2014, and his Msc. degree in computer science from the Federal University of Pernambuco, Brazil, in 2011. He is currently a postdoctoral researcher at the School of Electrical and Computer Engineering, Unicamp. His main research interests are multi-objective optimization, time series forecasting, general artificial intelligence, and machine learning. He is a student member of IEEE since 2009, a founding member and ex-Chair of the Student Chapter of the IEEE Computational Intelligence Society (CIS) at Unicamp, and the vice-Chair of the IEEE CIS Student Activities Subcommittee.

Fernando J. Von Zuben received his Dr.E.E. degree from the University of Campinas (Unicamp), Campinas, SP, Brazil, in 1996. He is currently the header of the Laboratory of Bioinformatics and Bio-inspired Computing (LBiC), and Full Professor at the Department of Computer Engineering and Industrial Automation, School of Electrical and Computer Engineering, Unicamp. The main topics of his research are computational intelligence, bio-inspired computing, multivariate data analysis, and machine learning. He coordinates open-ended research projects in these topics, tackling real-world problems in the areas of information technology, decision making, pattern recognition, and discrete and continuous optimization.